
Group 4

**SmartSpend AI
Software Architecture Document**

Version 2.0

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Revision History

Date	Version	Description	Author
17/07/2026	1.0	Update introduction, Architectural Goals and Constraints	Tạ Bình Phong
25/07/2026	1.1	Update logical view	Nguyễn Trịnh Tuấn Văn
25/07/2026	1.2	Update database component	Phan Văn Bá Đạt
26/7/2026	1.3	Update use case model and class diagram	Nguyễn Trịnh Tuấn Văn
26/07/2026	1.4	Update database component	Phan Văn Bá Đạt
09/08/2026	2.0	Update Deployment and Logical View	Tạ Bình Phong

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Software Architecture Document

1. Introduction

The Software Architecture Document provides a comprehensive overview of the system's architectural design. It serves as a blueprint for the development team, stakeholders, and future maintainers, offering insights into the high-level structure, major components, their interactions, and key architectural decisions.

1.1 Purpose

The purpose of this document is to describe the software architecture of the SmartSpend AI system. It aims to ensure that the 4-member development team and stakeholders have a shared understanding of the system architecture. This document will support decision-making during development, guide implementation, and facilitate communication among team members.

1.2 Scope

This Software Architecture Document outlines the overall architecture of the SmartSpend AI mobile application, focusing on its major components, modules, interfaces, and the way they interact. It explains the structural composition of the system—including the integration of OCR technology and AI chatbots—the core design principles, and the rationale behind the selection of architectural styles and patterns.

1.3 Overview

The following sections provide a comprehensive breakdown of the architecture of the SmartSpend AI system. The document is organized as follows:

- **Introduction:** Provides an introduction to the Software Architecture Document, its purpose, scope and an overview of the content.
- **Architectural Goals and Constraints:** Identifies the key goals and limitations that shaped architectural decisions.
- **Use-Case Model:** Presents the architectural perspective from the standpoint of system use cases and scenarios.
- **Logical View:** Details the logical organization of the system, emphasizing the division of responsibilities among different components.
- **Deployment:** Demonstrates how software components are deployed onto the physical hardware environment.
- **Implementation View:** Describes how the components are implemented and organized in the codebase.

2. Architectural Goals and Constraints

This section describes the primary goals and constraints that significantly impact the architectural design of the SmartSpend AI system. These are derived from the non-functional requirements and the project's development environment

2.1 Architectural Goals

- **Efficiency & Performance:**
 - Respond to standard user actions (e.g., login, manual entry, filtering, chart rendering) within 1.5 seconds in 95% of cases.
 - Execute the cumulative smart AI pipeline (receipt/screenshot OCR scanning and auto-categorization) within 8.0 seconds.
 - Maintain absolute stability and responsiveness during high-traffic hours, such as end-of-day periods or holiday seasons.
- **Availability & Reliability:**
 - Provide meaningful, user-friendly error messages and automatically redirect to the manual input form to recover gracefully if third-party APIs disconnect or system errors occur.
 - Automatically restart critical backend services and AI classification pipelines if they fail.
 - Support offline operational reliability by utilizing local data caching for manual entries, automatically synchronizing to the cloud database once internet connectivity is restored.
- **Security:**

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- Implement Role-Based Access Control (RBAC) to dynamically isolate individual consumer profile boundaries and securely unlock hidden administrative features for the Single Admin Account.
- Store passwords in the database using strong, secure one-way hashing algorithms.
- Encrypt sensitive financial records and personal data in transit via HTTPS and at rest within the database system.
- **Usability & Compatibility:**
 - Ensure the mobile application interface is fully responsive and optimized for different smartphone screen sizes and resolutions.
 - Support data portability by enabling users to seamlessly export historical transactional data into standard Excel or CSV formats.

2.2 Architectural Constraints

- **Development Limitations:** The project is a zero-budget academic endeavor bounded by a strict 12-week timeline across 5 Jira Sprints, executed by a fixed team of 4 members.
- **Deployment Environment:** The front-end user client must deploy natively and function correctly on the Android mobile operating system, with iOS deployment if possible. The administrative control panel utilizes a web-based Dashboard accessed via modern browsers (Chrome, Edge).
- **Third-Party Integrations:** The application relies on external third-party services and smart cloud APIs for OCR and LLM models, which must fully support the target Android ecosystem.
- **Development Tools:** Jira is used for sprint management, GitHub for source control and version management, Google Drive for document storage, and Figma for high-fidelity UI/UX prototyping.

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3. Use-Case Model



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4. Logical View

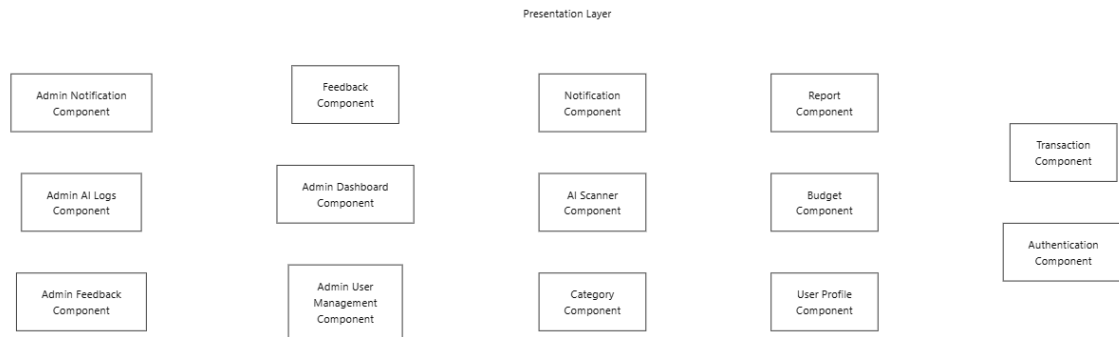
The SmartSpend AI system follows a 3-tier decoupled architecture, consisting of:

- **Presentation Layer:** Built using Expo/React Native (TypeScript), this layer renders the mobile user interface and communicates with the backend via RESTful APIs.
- **Business Logic Layer:** Implemented with Express.js, this layer manages routing, authentication, core business rules, and database interactions.
- **Data Layer:** Hosted on Supabase, which provides a cloud-based PostgreSQL database and integrated authentication and storage services.

This separation of concerns allows for high scalability, independent deployment of frontend/backend, easier testing, and potential replacement of individual layers without breaking the entire system.

4.1 Presentation Layer

Presentation Layer components are UI classes. Each class contains only UI attributes: pages, forms, buttons, panels, dialogs, charts, and other visual elements.

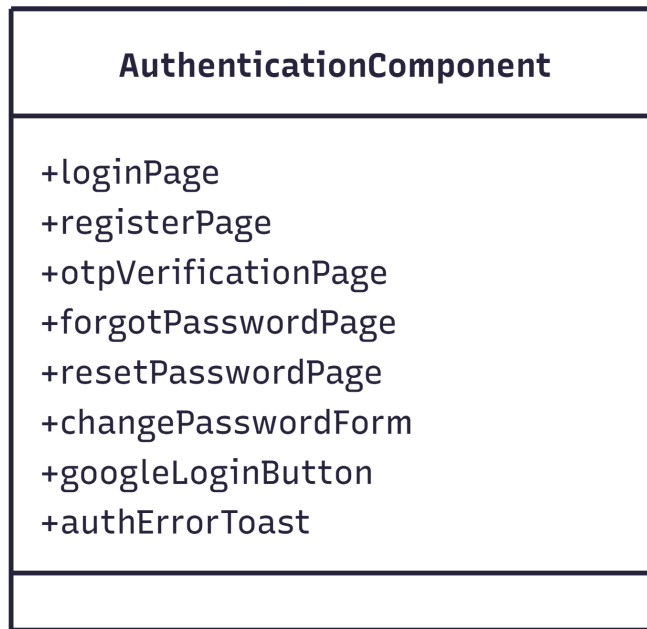


4.1.1 Component: Authentication Component

Layer: Presentation

Responsibility: Displays authentication screens and captures authentication-related input.

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Attribute	Description
loginPage	Login screen where user enters email and password.
registerPage	Registration screen where user enters name, email, and password.
otpVerificationPage	Screen for entering OTP during registration or password reset.
forgotPasswordPage	Screen for submitting email to start password recovery.
resetPasswordPage	Screen for entering a new password after OTP verification.
changePasswordForm	Form for logged-in users to change password.
googleLoginButton	UI button that starts Google login flow.

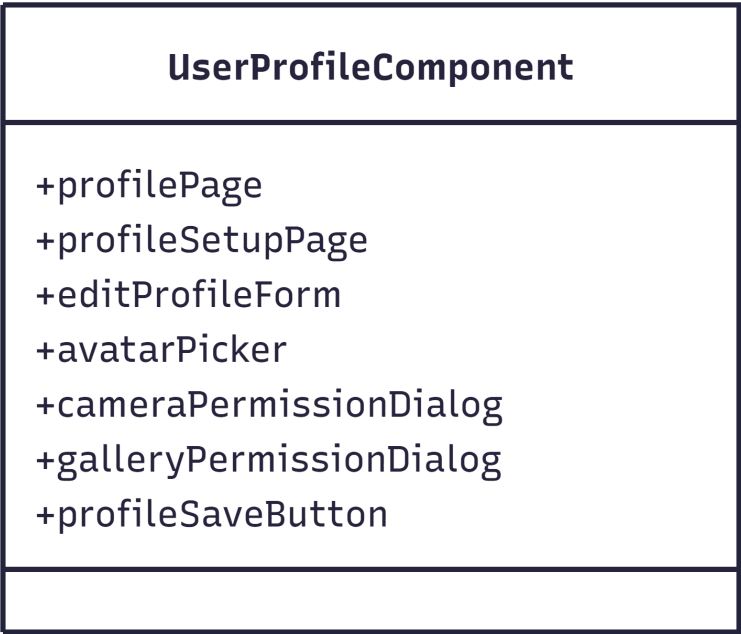
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authErrorToast	Toast area for authentication validation and error messages.
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4.1.2 Component: User Profile Component

Layer: Presentation

Responsibility: Displays profile view and profile editing elements.



Attribute	Description
profilePage	Page that shows current user profile information.
profileSetupPage	Initial profile setup page shown after successful registration.
editProfileForm	Form for editing full name, birthday, job, and income.
avatarPicker	UI element for choosing avatar from camera or gallery.
cameraPermissionDialog	Dialog shown when camera permission is required.

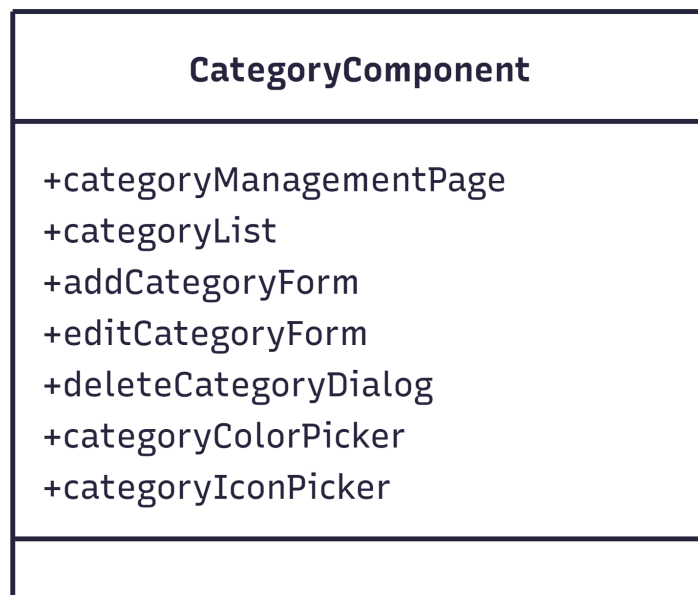
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galleryPermissionDialog	Dialog shown when gallery permission is required.
profileSaveButton	Button used to submit profile changes.

4.1.3 Component: Category Component

Layer: Presentation

Responsibility: Displays category list and category CRUD forms.



Attribute	Description
categoryManagementPage	Main screen for category management.
categoryList	List of default and custom categories.
addCategoryForm	Form for creating a new custom category.
editCategoryForm	Form for updating category name, color, or icon.
deleteCategoryDialog	Confirmation dialog shown before deleting a category.

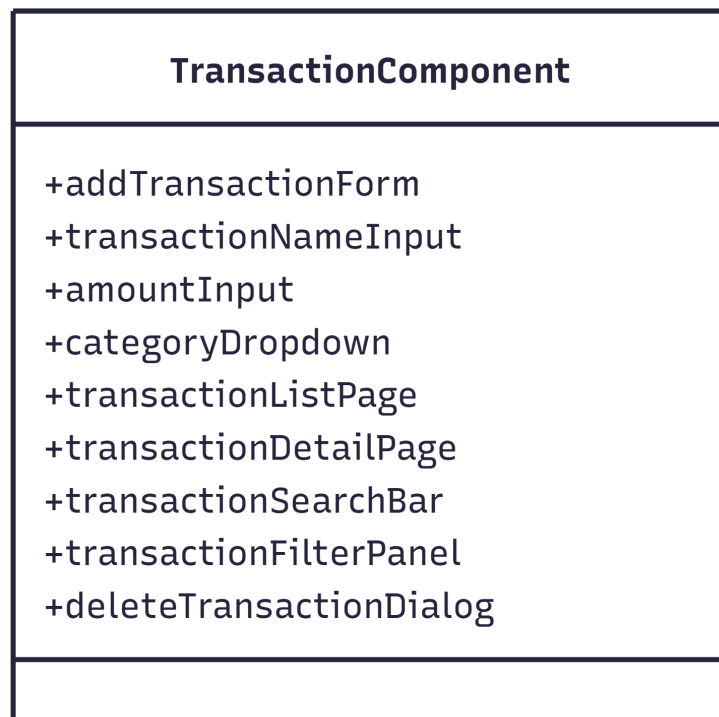
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categoryColorPicker	UI picker for selecting category color.
categoryIconPicker	UI picker for selecting category icon.

4.1.4 Component: Transaction Component

Layer: Presentation

Responsibility: Displays manual transaction entry, transaction detail, and transaction history UI.



Attribute	Description
addTransactionForm	Form for manually creating a transaction.
transactionNameInput	Input field for transaction name.
amountInput	Input field for transaction amount.
categoryDropdown	Selection list for transaction category.

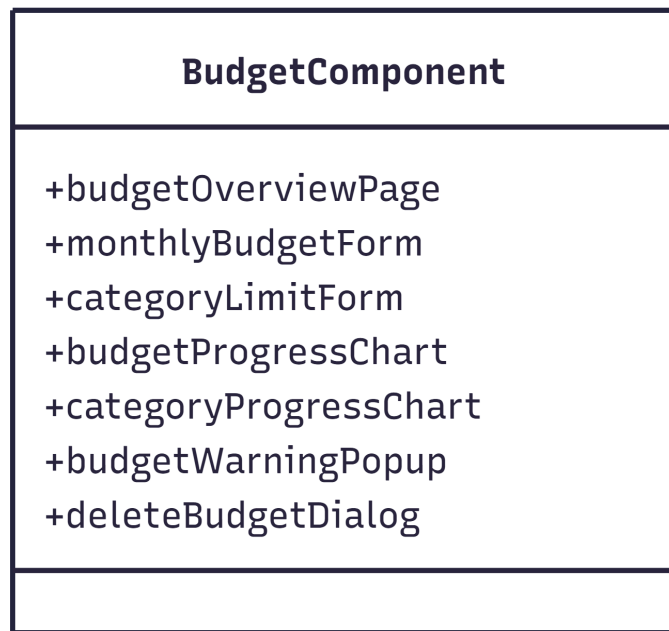
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transactionListPage	Page showing all transactions.
transactionDetailPage	Page showing details of one transaction.
transactionSearchBar	Search input used for keyword search.
transactionFilterPanel	Panel containing date, type, amount, and category filters.
deleteTransactionDialog	Confirmation dialog shown before deleting a transaction.

4.1.5 Component: Budget Component

Layer: Presentation

Responsibility: Displays monthly budget, category limits, budget charts, and warning messages.



Attribute	Description
budgetOverviewPage	Main budget management screen.

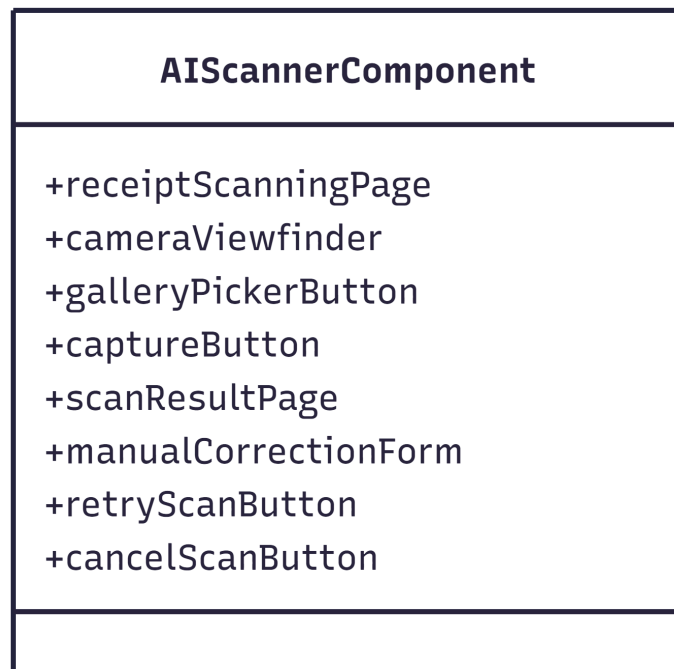
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monthlyBudgetForm	Form for setting or updating total monthly budget.
categoryLimitForm	Form for setting spending limits by category.
budgetProgressChart	Chart showing total spending progress against monthly budget.
categoryProgressChart	Chart showing category-level spending progress.
budgetWarningPopup	Popup shown when spending approaches or exceeds budget.
deleteBudgetDialog	Confirmation dialog shown before deleting a budget.

4.1.6 Component: AI Scanner Component

Layer: Presentation

Responsibility: Displays receipt scanning, image selection, result review, correction, and save/cancel UI.



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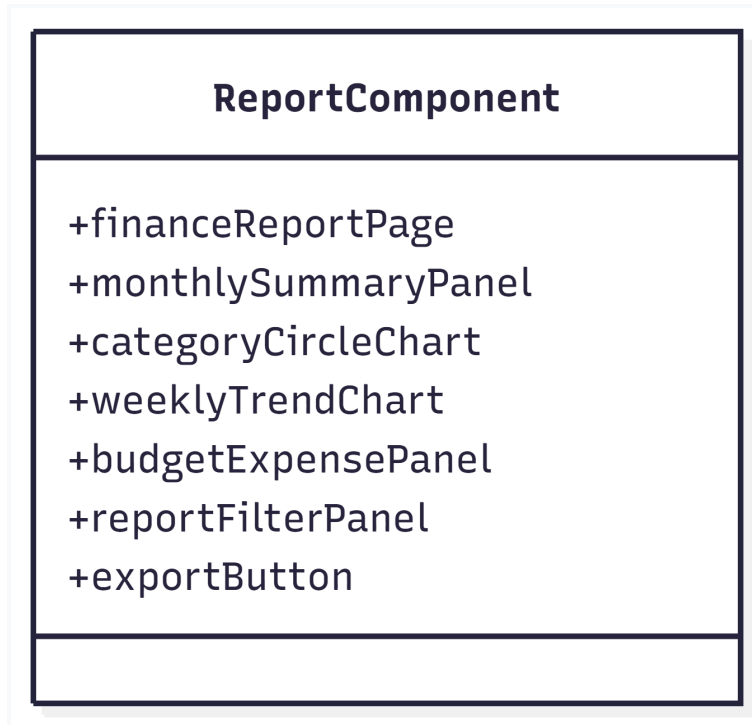
Attribute	Description
receiptScanningPage	Screen used to scan receipts.
cameraViewfinder	Camera preview region for receipt capture.
galleryPickerButton	Button for selecting receipt image from gallery.
captureButton	Button for taking receipt photo.
scanResultPage	Page showing OCR/AI extracted result.
manualCorrectionForm	Form for correcting wrong AI result fields.
retryScanButton	Button used when OCR/AI processing fails.
cancelScanButton	Button used to cancel scan flow.

4.1.7 Component: Report Component

Layer: Presentation

Responsibility: Displays financial reports, charts, and export controls.

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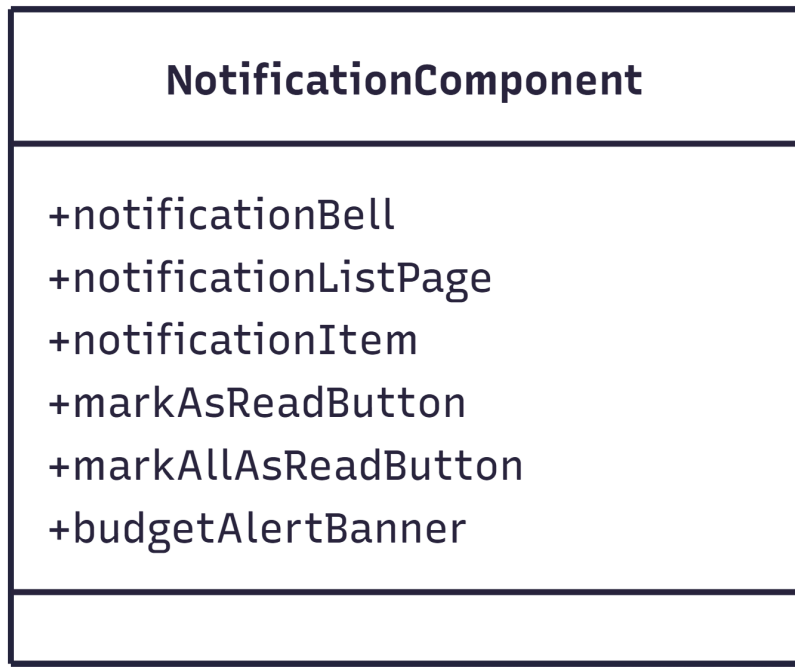
Attribute	Description
financeReportPage	Main financial report screen.
monthlySummaryPanel	Panel showing monthly income, expense, and savings.
categoryCircleChart	Circle chart showing expenses by category.
weeklyTrendChart	Multi-line chart showing category expenses by week.
budgetExpensePanel	Panel comparing budget and current month expense.
reportFilterPanel	Filter panel for report period or category selection.
exportButton	Button for exporting transactions or reports.

4.1.8 Component: Notification Component

Layer: Presentation

Responsibility: Displays user notification history and notification read state.

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Attribute	Description
notificationBell	Header icon showing notification count.
notificationListPage	Page listing user notifications.
notificationItem	UI row representing one notification.
markAsReadButton	Button for marking one notification as read.
markAllAsReadButton	Button for marking all notifications as read.
budgetAlertBanner	Banner shown for budget warning notifications.

4.1.9 Feedback Component

Layer: Presentation

Responsibility: Displays user feedback submission and feedback history UI.

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FeedbackPresentationComponent

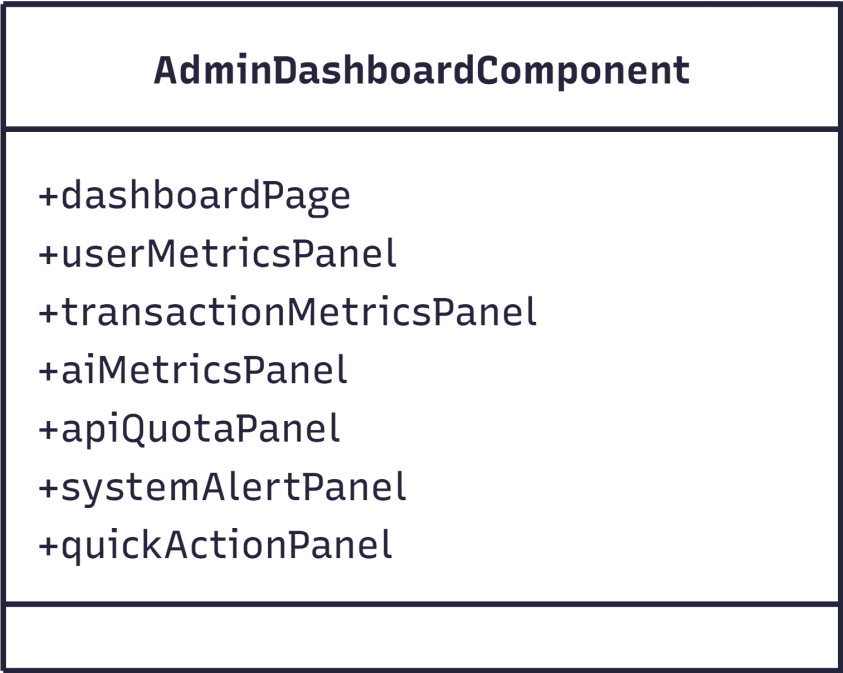
+feedbackForm
 +feedbackContentInput
 +feedbackAttachmentPicker
 +myFeedbackListPage
 +feedbackDetailPage
 +submitFeedbackButton

Attribute	Description
feedbackForm	Form for submitting new feedback.
feedbackContentInput	Text input for feedback content.
feedbackAttachmentPicker	UI element for attaching screenshots or files.
myFeedbackListPage	Page showing feedback submitted by current user.
feedbackDetailPage	Page showing feedback status and admin response.
submitFeedbackButton	Button used to submit feedback.

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4.1.10 Component: Admin Dashboard Component

- **Layer:** Presentation
- **Responsibility:** Displays admin dashboard metrics, alerts, and quick-action navigation.



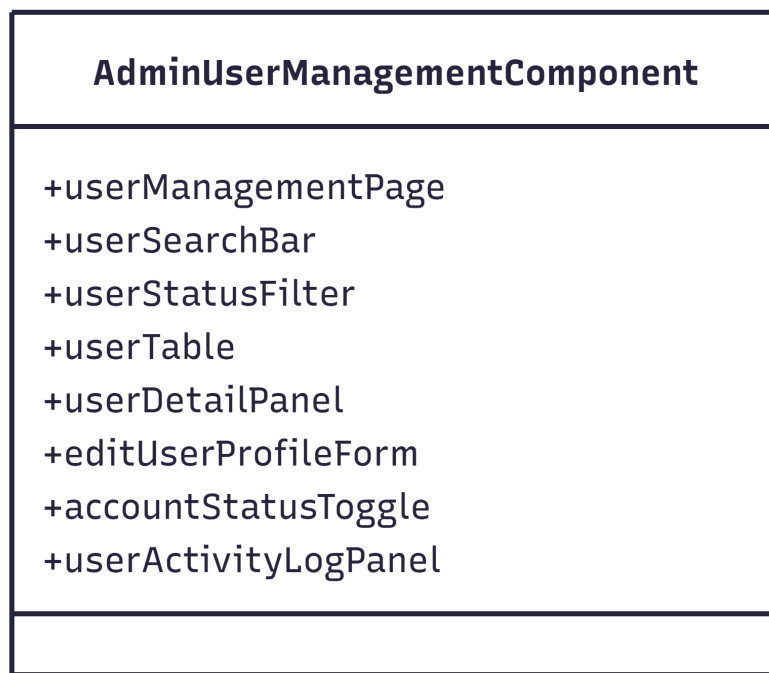
Attribute	Description
dashboardPage	Main admin dashboard page.
userMetricsPanel	Panel showing user counts and trends.
transactionMetricsPanel	Panel showing transaction statistics.
aiMetricsPanel	Panel showing AI scan success/failure metrics.

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apiQuotaPanel	Panel showing OCR/AI API cost and quota.
systemAlertPanel	Panel showing pending feedbacks or unresolved issues.
quickActionPanel	Panel linking to user management, notification center, AI logs, and feedback pages.

4.1.11 Component: Component: Admin User Management Component

- **Layer:** Presentation
- **Responsibility:** Displays user management screens for admin.



Attribute	Description
userManagementPage	Page for managing registered users.

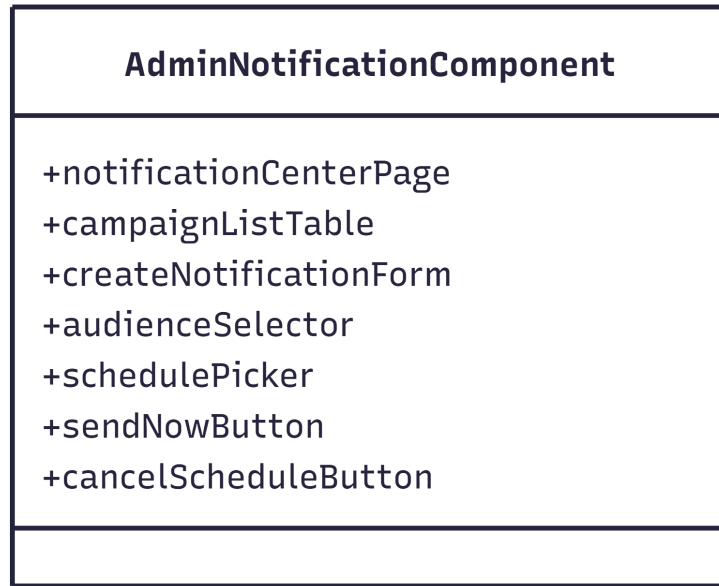
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userSearchBar	Search input for name or email.
userStatusFilter	Filter for active or inactive accounts.
userTable	Table listing user accounts.
userDetailPanel	Panel showing full user profile details.
editUserProfileForm	Admin form for editing user profile information.
accountStatusToggle	UI control for activating or deactivating accounts.
userActivityLogPanel	Panel showing user activity logs.

4.1.12 Component: Admin Notification Component

- **Layer:** Presentation
- **Responsibility:** Displays notification campaign management UI.

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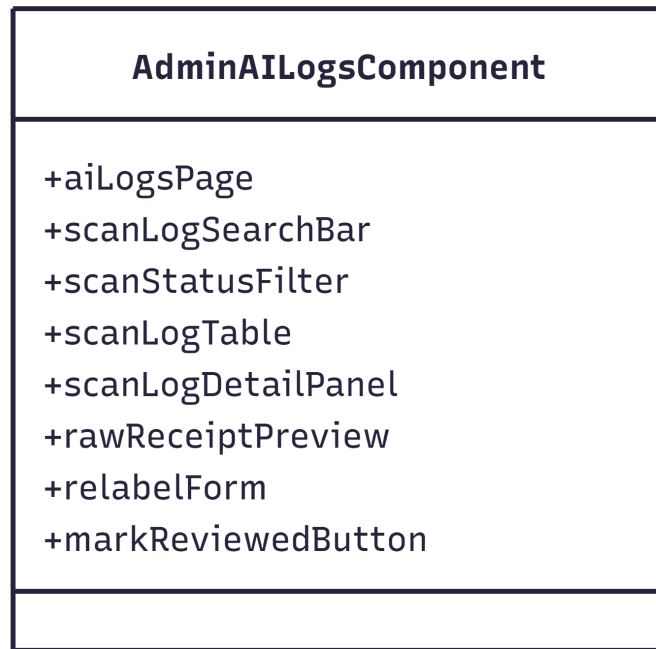
Attribute	Description
notificationCenterPage	Admin page for managing push campaigns.
campaignListTable	Table showing past and scheduled campaigns.
createNotificationForm	Form for writing title and message content.
audienceSelector	Selector for choosing target audience.
schedulePicker	UI element for choosing immediate or scheduled delivery.
sendNowButton	Button for immediate broadcast.

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cancelScheduleButton	Button for cancelling scheduled campaign.
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4.1.13 Component: Admin AI Logs Component

- **Layer:** Presentation
- **Responsibility:** Displays AI scan logs and relabeling UI for admin.



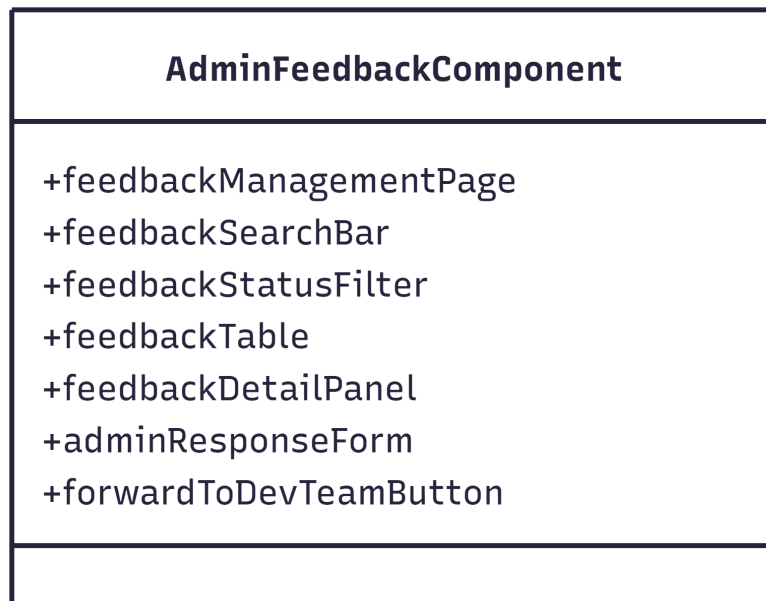
Attribute	Description
aiLogsPage	Admin page for AI scan log review.
scanLogSearchBar	Search input for transaction ID or scan ID.
scanStatusFilter	Filter for success, failed, or reviewed scans.

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scanLogTable	Table listing AI scan logs.
scanLogDetailPanel	Panel showing detailed OCR and AI result.
rawReceiptPreview	Preview of uploaded receipt image.
relabelForm	Form for correcting AI category classification.
markReviewedButton	Button for marking a scan log as reviewed.

4.1.14 Component: Admin Feedback Component

- **Layer:** Presentation
- **Responsibility:** Displays feedback management screens for admin.



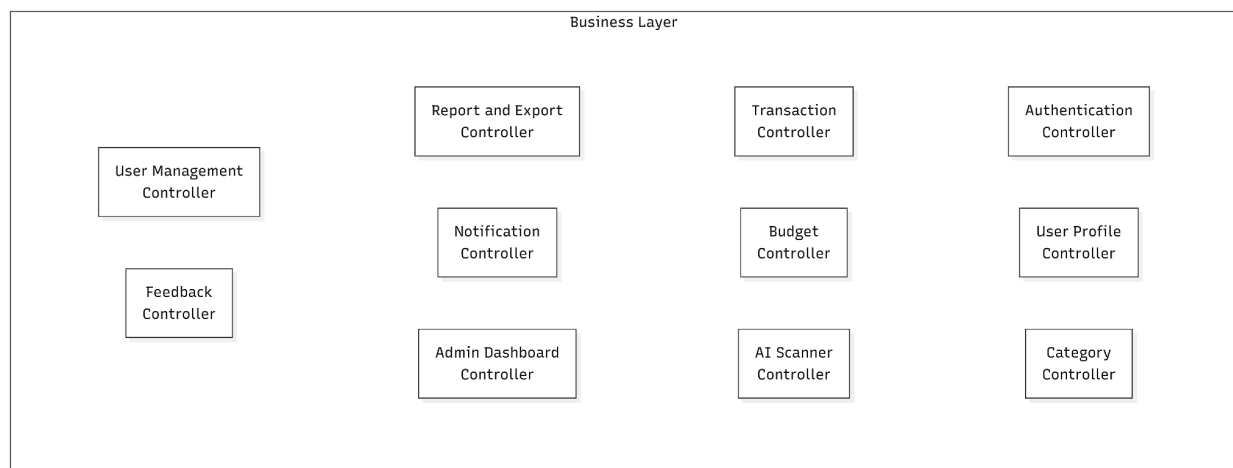
Attribute	Description

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feedbackManagementPage	Admin page for managing feedback.
feedbackSearchBar	Search input for feedback keyword filtering.
feedbackStatusFilter	Filter for pending, in-progress, or resolved feedback.
feedbackTable	Table showing feedback entries.
feedbackDetailPanel	Panel showing feedback details and attachments.
adminResponseForm	Form for writing admin response.
forwardToDevTeamButton	Button for flagging feedback as a developer issue.

4.2 Business Layer

Business Layer components are controller classes. Each controller contains methods that execute the business functions requested by Presentation Layer components.

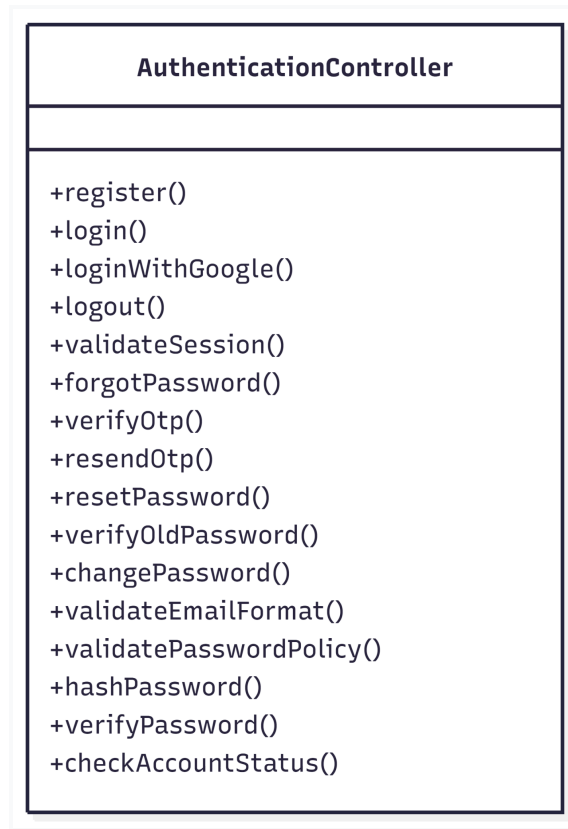


4.2.1 Component: Authentication Controller

Layer: Business

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Responsibility: Executes registration, login, OTP, password reset, session, and password-change workflows.



Method	Description
register	Creates a new account, validates email/password, stores hashed password, and starts OTP verification.
login	Authenticates user credentials and creates a session.
loginWithGoogle	Authenticates with Google OAuth and creates a session.
logout	Invalidates current user session.

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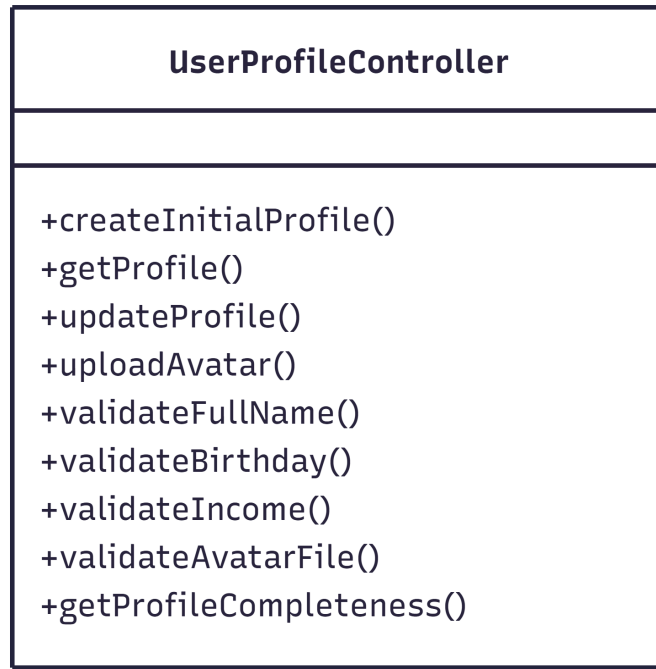
validateSession	Checks whether session token is still valid.
forgotPassword	Starts password recovery by sending OTP to email.
verifyOtp	Validates OTP for registration or password reset.
resendOtp	Sends a new OTP when old OTP expires.
resetPassword	Saves new password after OTP verification.
verifyOldPassword	Checks old password before change password.
changePassword	Updates password after verifying old password and policy.
validateEmailFormat	Rejects invalid email format.
validatePasswordPolicy	Checks password length, number, uppercase, and special character rules.
hashPassword	Creates secure one-way password hash.
verifyPassword	Compare entered password with stored hash.
checkAccountStatus	Blocks inactive or disabled accounts from secure workflows.

4.2.2 Component: User Profile Controller

Layer: Business

Responsibility: Executes initial profile setup and profile update workflows.

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Method	Description
createInitialProfile	Saves first profile after registration OTP success.
getProfile	Retrieves current profile data.
updateProfile	Updates profile fields after validation.
uploadAvatar	Stores profile image and returns avatar URL.
validateFullName	Rejects blank or too long names.
validateBirthday	Rejects future or unrealistic birthday.

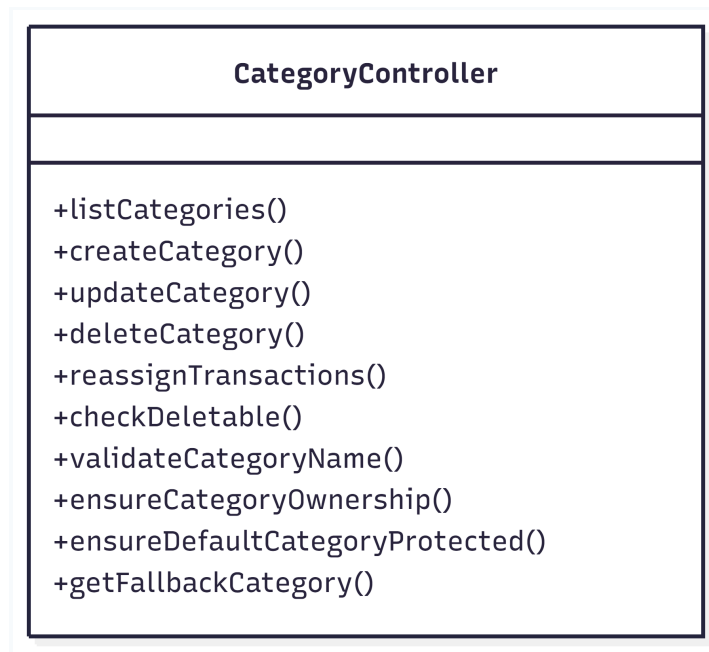
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Method	Description
validateIncome	Rejects non-numeric or negative income.
validateAvatarFile	Checks avatar type and file size.
getProfileCompleteness	Calculates percentage of completed profile fields.

4.2.3 Component: Category Controller

Layer: Business

Responsibility: Executes category creation, update, deletion, uniqueness validation, ownership checks, and transaction reassignment.



Method	Description
listCategories	Returns default and custom categories.

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Method	Description
createCategory	Creates a custom category after validation.
updateCategory	Updates category name, icon, or color.
deleteCategory	Deletes category after Presentation Layer confirmation.
reassignTransactions	Moves transactions to fallback category before deletion.
checkDeletable	Checks whether category dependencies allow deletion.
validateCategoryName	Checks blank and duplicate category names.
ensureCategoryOwnership	Ensures user owns the category.
ensureDefaultCategoryProtected	Prevents deleting protected default categories.
getFallbackCategory	Returns Uncategorized/Other category for reassignment.

4.2.4 Component: Transaction Controller

Layer: Business

Responsibility: Executes manual transaction creation, transaction detail, edit/delete, search/filter/history, and balance recalculation.

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Method	Description
createTransaction	Saves manual transaction or confirmed AI scan transaction.
validateTransaction	Validates complete transaction command.
validateTransactionName	Ensures transaction name is provided.

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Method	Description
validateAmount	Ensures amount is numeric and greater than zero.
validateDate	Ensures date/time is valid.
validateCategoryExists	Ensures selected category exists.
ensureTransactionOwnership	Prevents user from editing another user's transaction.
calculateBalance	Calculates current wallet balance.
calculateBalanceByPeriod	Calculates balance for a selected date range.
getTransaction	Retrieves one transaction detail.
updateTransaction	Updates transaction and recalculates related data.
deleteTransaction	Deletes transaction and recalculates balance.
listTransactions	Returns paginated transaction list.
searchTransactions	Searches by transaction name, note, or keyword.
filterTransactions	Filters by date, type, category, or amount.
getTransactionHistory	Returns chronological transaction history.

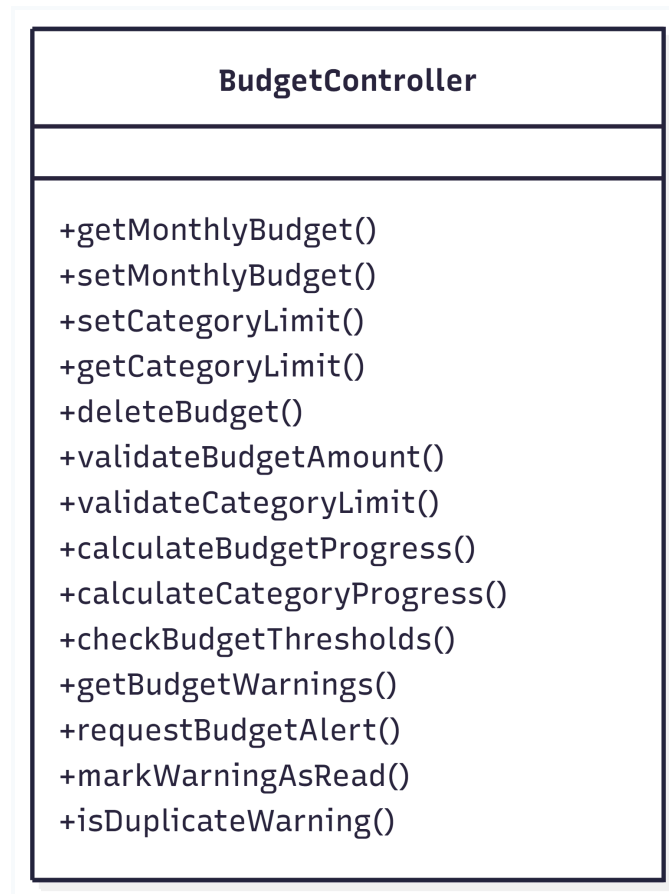
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Method	Description
getTransactionSummary	Returns income and expense totals for a period.
recalculateAfterChange	Refreshes balance, budget warning, and report inputs after transaction changes.

4.2.5 Component: Budget Controller

Layer: Business

Responsibility: Executes monthly budget setup, category limits, progress calculation, threshold detection, and budget warning requests.



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Method	Description
getMonthlyBudget	Retrieves monthly budget and progress.
setMonthlyBudget	Creates or updates total monthly budget.
setCategoryLimit	Sets spending limit for one category.
getCategoryLimit	Retrieves category limit.
deleteBudget	Deletes budget after Presentation Layer confirmation.
validateBudgetAmount	Rejects invalid monthly budget amount.
validateCategoryLimit	Rejects invalid category limit amount.
calculateBudgetProgress	Calculates used and remaining monthly budget.
calculateCategoryProgress	Calculates spending progress by category.
checkBudgetThresholds	Checks whether expense transaction triggers warning.
getBudgetWarnings	Retrieves active warning messages.
requestBudgetAlert	Requests Notification Controller to deliver alert.
markWarningAsRead	Marks warning as acknowledged.

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Method	Description
isDuplicateWarning	Prevents repeated alerts for same threshold.

4.2.6 Component: AI Scanner Controller

Layer: Business

Responsibility: Executes receipt upload, OCR/AI processing, result correction, transaction confirmation, and admin scan log management.

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Method	Description
uploadReceipt	Accepts receipt image from camera or gallery.

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Method	Description
validateImagePermission	Handles missing camera/gallery permission state.
validateImageFile	Checks file type and size.
validateImageQuality	Detects blurry or unrecognizable image.
processReceiptImage	Runs OCR/AI processing.
extractReceiptData	Extracts amount, date, merchant, and text data.
suggestCategory	Suggests category from receipt data.
getScanResult	Retrieves current scan result for review.
saveManualCorrection	Stores user-corrected AI output before saving.
confirmTransaction	Saves confirmed scan result through Transaction Controller.
rejectTransaction	Cancels scan result without saving transaction.
retryScanProcessing	Retries OCR/AI when network or provider fails.
clearTemporaryScanData	Clears temporary image/result data after cancel.
recordScanLog	Stores scan success/failure log.

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Method	Description
listScanLogs	Returns scan log list for admin.
getScanLogDetail	Returns OCR text, image, category, and error detail.
getScanStatistics	Returns AI success rate and processing metrics.
exportScanLogs	Exports scan logs for analysis.
getScanErrorDetails	Returns failure reason and error code.
relabelScanData	Saves corrected category label from admin.
markScanAsReviewed	Marks scan log as reviewed.

4.2.7 Component: Report and Export Controller

Layer: Business

Responsibility: Executes financial report aggregation, chart data generation, and export workflows.

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ReportAndExportController
<pre> +getMonthlyReport() +getCategoryBreakdown() +getExpenseChart() +getIncomeChart() +getBudgetExpenseOverview() +getWeeklyCategoryTrend() +getCurrentMonthExpense() +getReportFilterOptions() +calculateSavings() +exportTransactions() +downloadExport() +getExportStatus() </pre>

Method	Description
getMonthlyReport	Returns income, expense, and savings summary.
getCategoryBreakdown	Groups expenses by category.
getExpenseChart	Returns expense chart data.
getIncomeChart	Returns income chart data.
getBudgetExpenseOverview	Compares current budget and actual expense.
getWeeklyCategoryTrend	Returns multi-line chart data by category/week.
getCurrentMonthExpense	Computes current month expense.

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Method	Description
getReportFilterOptions	Returns available filter options.
calculateSavings	Computes income minus expense.
exportTransactions	Creates export file from transaction data.
downloadExport	Returns generated export file.
getExportStatus	Checks export job state.

4.2.8 Component: Notification Controller

Layer: Business

Responsibility: Executes user notification delivery, budget alerts, push tokens, and admin notification campaigns.

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NotificationController
<pre> +registerPushToken() +listUserNotifications() +markAsRead() +markAllAsRead() +sendToUser() +sendBudgetAlert() +validateNotificationContent() +validateAudience() +validateSchedule() +createCampaign() +getCampaignDetail() +listCampaigns() +scheduleNotification() +cancelScheduledNotification() +sendImmediateNotification() +retryFailedCampaign() +sendCampaign() +updateNotificationStatus() </pre>

Method	Description
registerPushToken	Stores device push token.
listUserNotifications	Lists notification history for user.
markAsRead	Marks one notification as read.
markAllAsRead	Marks all notifications as read.
sendToUser	Sends one notification through FCM.

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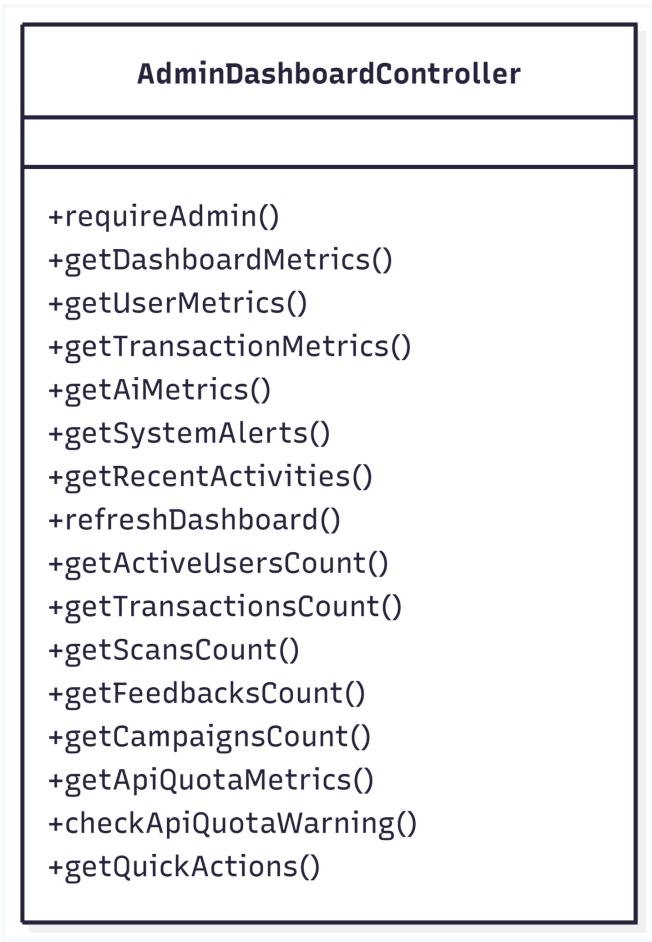
Method	Description
sendBudgetAlert	Sends budget warning notification.
validateNotificationContent	Checks missing title or message.
validateAudience	Checks selected target audience.
validateSchedule	Checks delivery time.
createCampaign	Creates admin notification campaign.
getCampaignDetail	Returns campaign detail.
listCampaigns	Lists campaigns.
scheduleNotification	Schedules campaign for later.
cancelScheduledNotification	Cancels scheduled campaign.
sendImmediateNotification	Sends campaign immediately.
retryFailedCampaign	Retries failed FCM broadcast.
sendCampaign	Executes campaign delivery.
updateNotificationStatus	Updates campaign status.

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4.2.9 Component: Admin Dashboard Controller

Layer: Business

Responsibility: Executes admin dashboard metrics, system alerts, quick actions, and API quota monitoring.



Method	Description
requireAdmin	Rejects non-admin users.
getDashboardMetrics	Returns full dashboard data.
getUserMetrics	Returns user growth and activity metrics.

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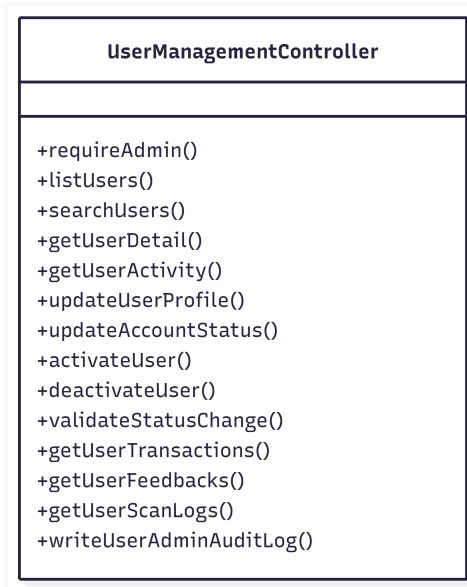
Method	Description
getTransactionMetrics	Returns transaction and manual-vs-AI metrics.
getAiMetrics	Returns AI success/failure and processing metrics.
getSystemAlerts	Returns pending feedback, bugs, and warnings.
getRecentActivities	Returns recent system activity timeline.
refreshDashboard	Reloads dashboard data.
getActiveUsersCount	Counts active users.
getTransactionsCount	Counts transactions in a period.
getScansCount	Counts AI scans in a period.
getFeedbacksCount	Counts feedback records.
getCampaignsCount	Counts notification campaigns.
getApiQuotaMetrics	Returns OCR/AI API cost and quota status.
checkApiQuotaWarning	Detects low quota warning.
getQuickActions	Returns dashboard shortcut entries.

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4.2.10 Component: User Management Controller

Layer: Business

Responsibility: Executes admin user listing, search, detail view, profile edit, account status update, and user activity lookup.



Method	Description
requireAdmin	Checks admin permission before user management.
listUsers	Returns paginated registered users.
searchUsers	Searches users by name, email, or status.
getUserDetail	Returns account and profile detail.
getUserActivity	Returns user activity history.
updateUserProfile	Allows admin to edit user profile information.

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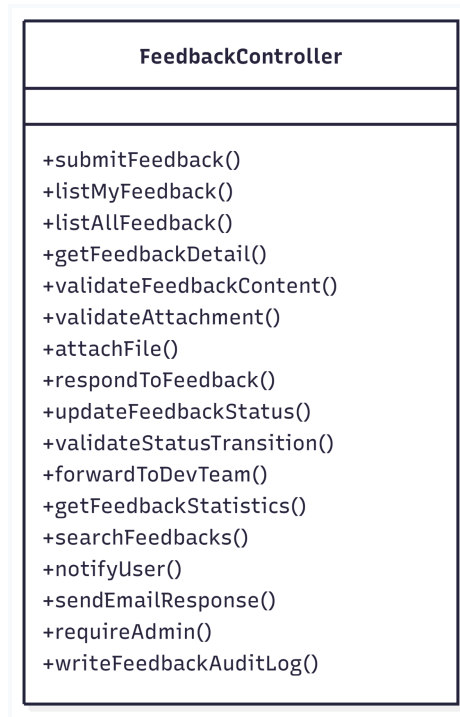
Method	Description
updateAccountStatus	Updates account active/inactive status.
activateUser	Restores login access.
deactivateUser	Blocks login access.
validateStatusChange	Prevents invalid account state transition.
getUserTransactions	Returns transactions of selected user.
getUserFeedbacks	Returns feedback submitted by selected user.
getUserScanLogs	Returns AI scan logs filtered by selected user.
writeUserAdminAuditLog	Records sensitive admin user action.

4.2.11 Component: Feedback Controller

Layer: Business

Responsibility: Executes user feedback submission, attachment storage, admin review, response, status update, forwarding, and user notification.

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Method	Description
submitFeedback	Creates user feedback.
listMyFeedback	Lists feedback submitted by current user.
listAllFeedback	Lists all feedback for admin.
getFeedbackDetail	Returns full feedback details and attachments.
validateFeedbackContent	Rejects blank or too long feedback.
validateAttachment	Checks attachment file type and size.

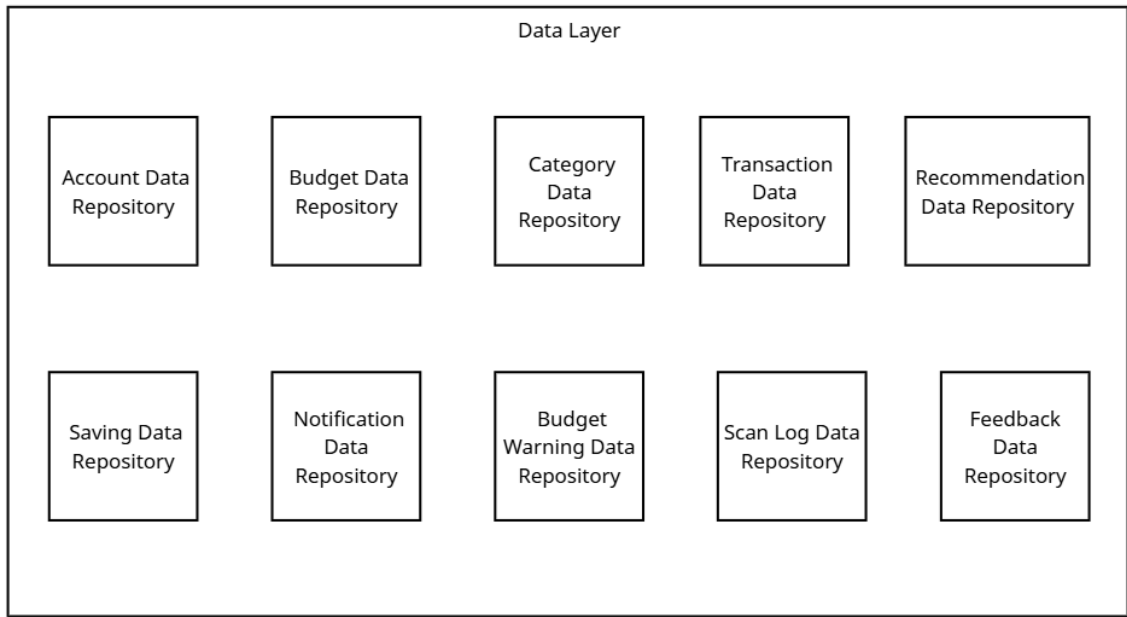
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Method	Description
attachFile	Stores attachment and returns URL.
respondToFeedback	Saves admin response.
updateFeedbackStatus	Changes feedback workflow status.
validateStatusTransition	Prevents invalid status transition.
forwardToDevTeam	Flags feedback as developer issue.
getFeedbackStatistics	Returns counts by category and status.
searchFeedbacks	Filters feedback by keyword or category.
notifyUser	Sends in-app notification after response.
sendEmailResponse	Sends automated email response.
requireAdmin	Protects admin feedback operations.
writeFeedbackAuditLog	Records admin feedback action.

4.3 Data layer

Data Layer components persist and retrieve data. They do not call Presentation or Business components.

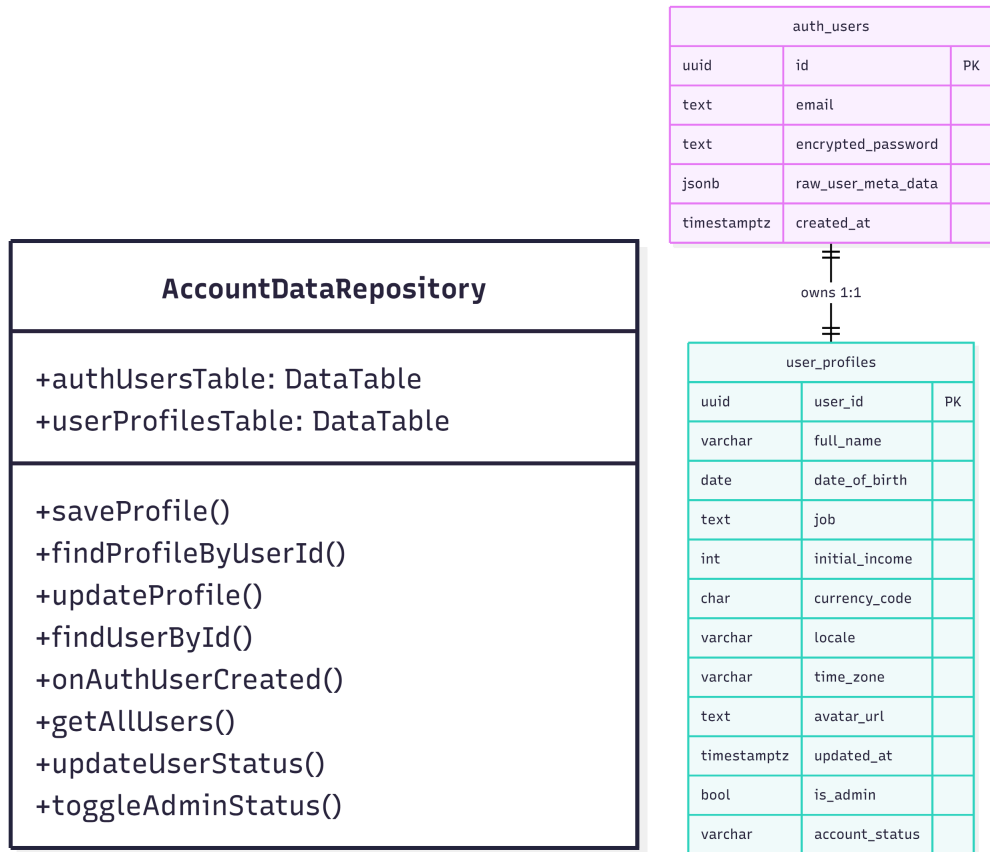
SmartSpend AI	Version: 1.4
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4.3.1 Component: Account Data Repository

Layer: Data

Responsibility: Stores user account identity from Supabase Auth and extended user profile information.



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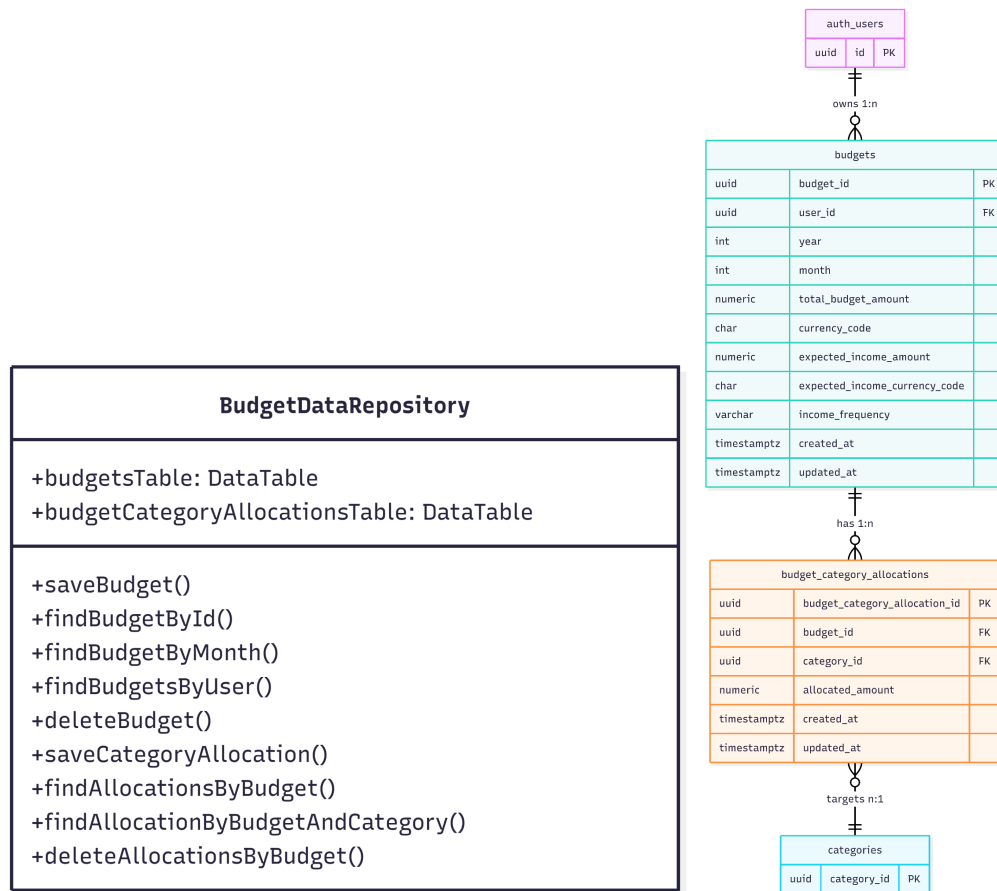
Method	Description
saveProfile	Inserts or updates user profile.
findProfileByUserId	Finds profile by user UUID.
updateProfile	Updates profile fields after validation.
findUserById	Finds user by UUID from auth.users.
onAuthUserCreated	Trigger function to auto-create profile on new user signup.
getAllUsers	Returns all user profiles (admin only).
updateUserStatus	Updates user account status.
toggleAdminStatus	Toggles admin status for a user.

4.3.2 Component: Budget Data Repository

Layer: Data

Responsibility: Stores monthly budgets and category-level spending allocations.

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Method	Description
saveBudget	Inserts or updates monthly budget.
findBudgetById	Finds budget by budget_id.
findBudgetByMonth	Finds budget by user, year and month
findBudgetsByUser	Returns all budgets for a user.
deleteBudget	Deletes budget and its allocations.

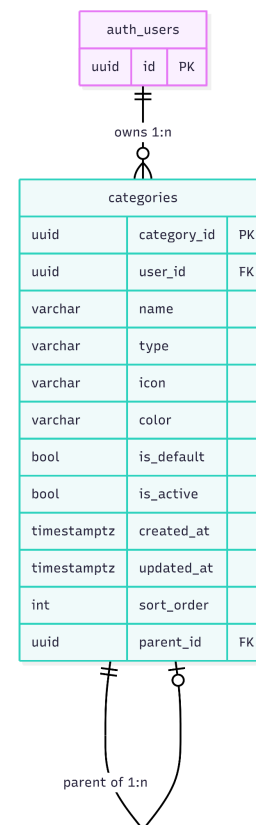
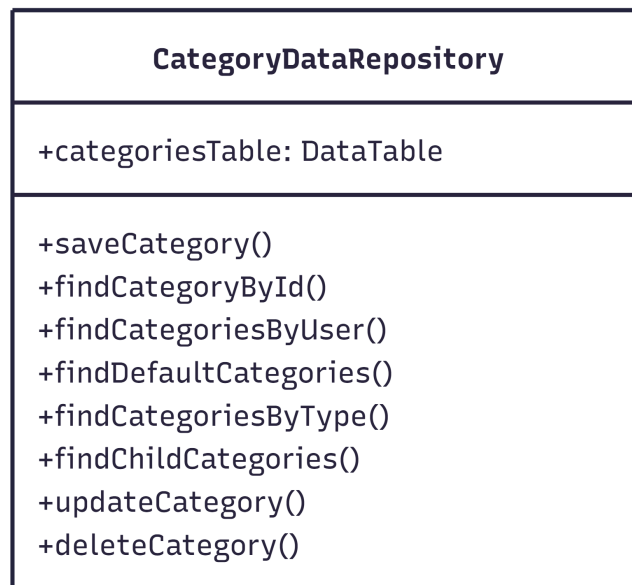
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Method	Description
saveCategoryAllocation	Inserts or updates category allocation.
findAllocationsByBudget	Returns all allocations for a budget.
findAllocationByBudgetAndCategory	Finds specific allocation by budget and category.
deleteAllocationsByBudget	Deletes all allocations for a budget.

4.3.3 Component:Category Data Repository

Layer: Data

Responsibility: Stores expense and income categories including default and custom categories with parent-child relationships.



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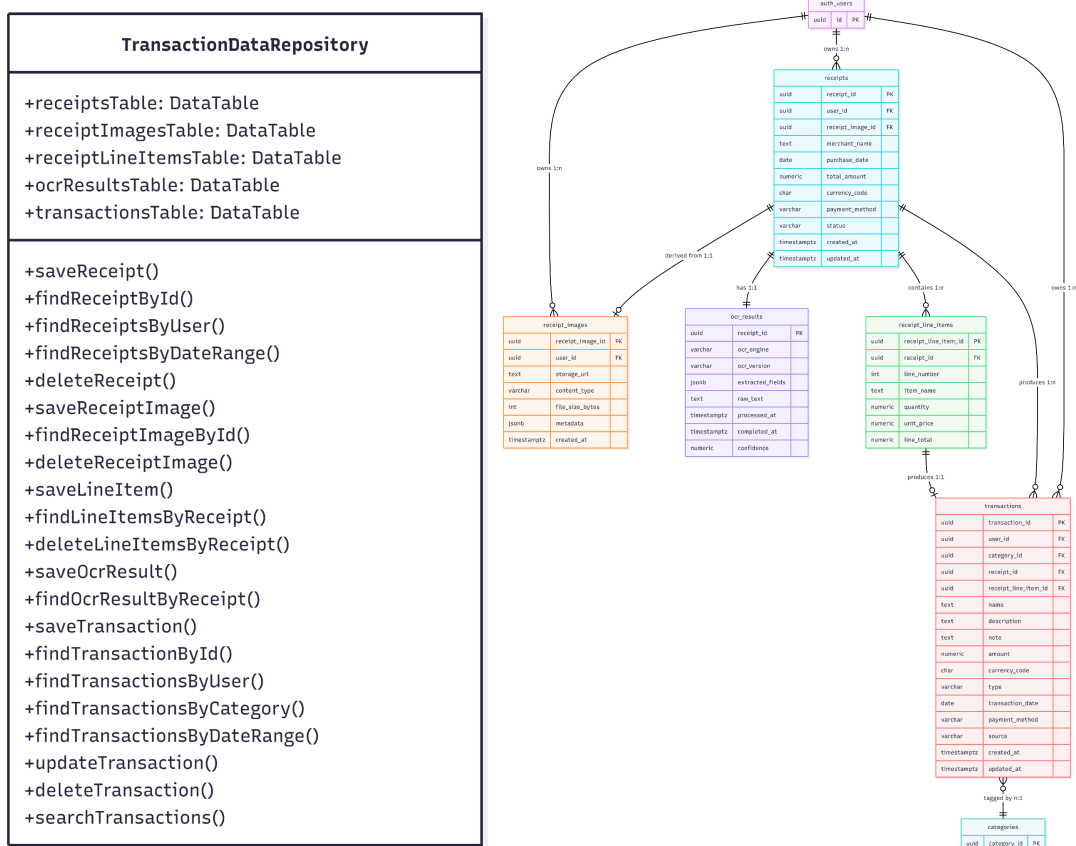
Method	Description
saveCategory	Inserts or updates category.
findCategoryById	Finds category by category_id.
findCategoriesByUser	Returns all categories for a user (including defaults).
findDefaultCategories	Returns all default categories (user_id IS NULL).
findCategoriesByType	Returns categories filtered by type (expense/income).
findChildCategories	Returns child categories of a parent category.
updateCategory	Updates category fields.
deleteCategory	Deletes category.

4.3.4 Component: Transaction Data Repository

Layer: Data

Responsibility: Stores receipts, receipt images, OCR results, line items, and transactions (income and expenses).

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Method	Description
saveReceipt	Inserts or updates receipt.
findReceiptById	Finds receipt by receipt_id.
findReceiptsByUser	Returns all receipts for a user.
findReceiptsByDateRange	Returns receipts within date range.
deleteReceipt	Deletes receipt and related line items.
saveReceiptImage	Saves receipt image metadata.

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Method	Description
findReceiptImageById	Finds receipt image by ID.
deleteReceiptImage	Deletes receipt image.
saveLineItem	Inserts or updates line item.
findLineItemsByReceipt	Returns all line items for a receipt.
deleteLineItemsByReceipt	Deletes all line items for a receipt.
saveOcrResult	Saves OCR processing result.
findOcrResultByReceipt	Finds OCR result by receipt ID.
saveTransaction	Inserts or updates transaction.
findTransactionById	Finds transaction by transaction_id.
findTransactionsByUser	Returns all transactions for a user.
findTransactionsByCategory	Returns transactions filtered by category.
findTransactionsByDateRange	Returns transactions within date range.
updateTransaction	Updates transaction fields.
deleteTransaction	Deletes transaction.

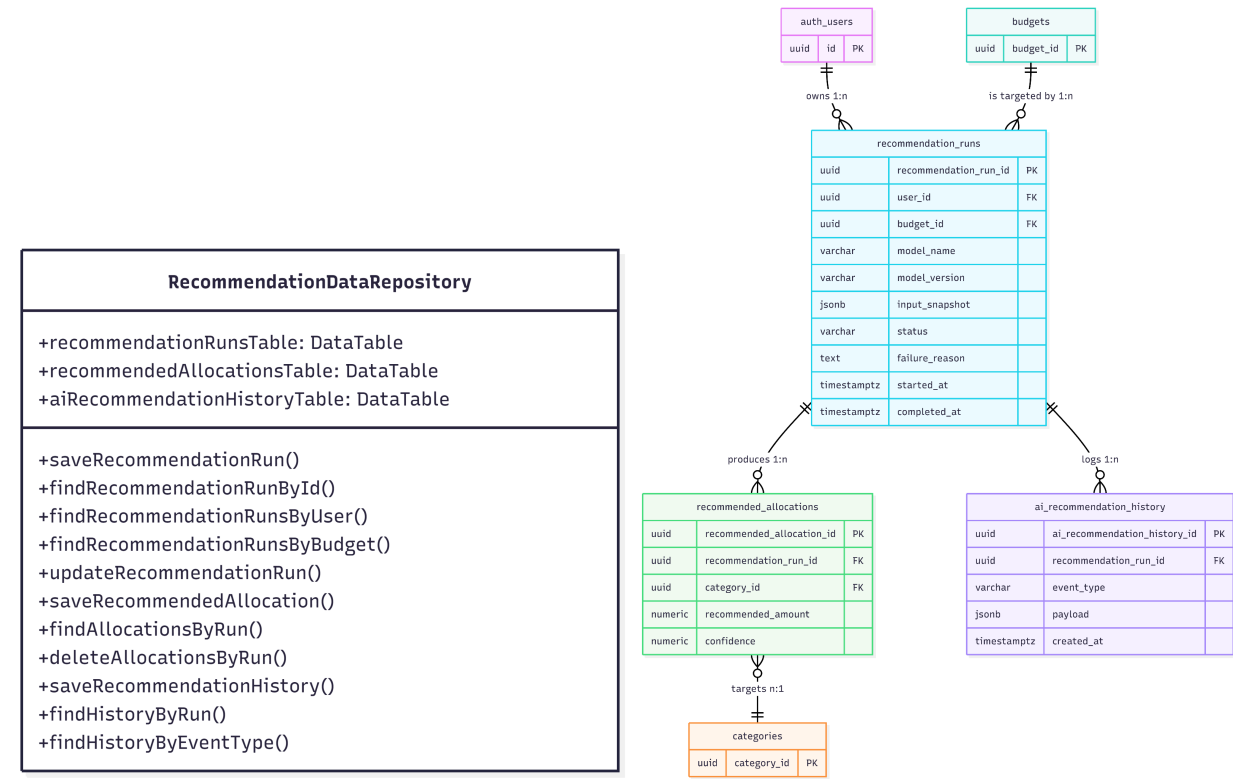
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Method	Description
searchTransactions	Searches transactions with filters and pagination.

4.3.5 Component: Recommendation Data Repository

Layer: Data

Responsibility: Stores AI recommendation runs, suggested allocations, and recommendation history.



Method	Description
saveRecommendationRun	Inserts or updates recommendation run.
findRecommendationRunById	Finds run by recommendation_run_id.
findRecommendationRunsByUser	Returns all runs for a user.

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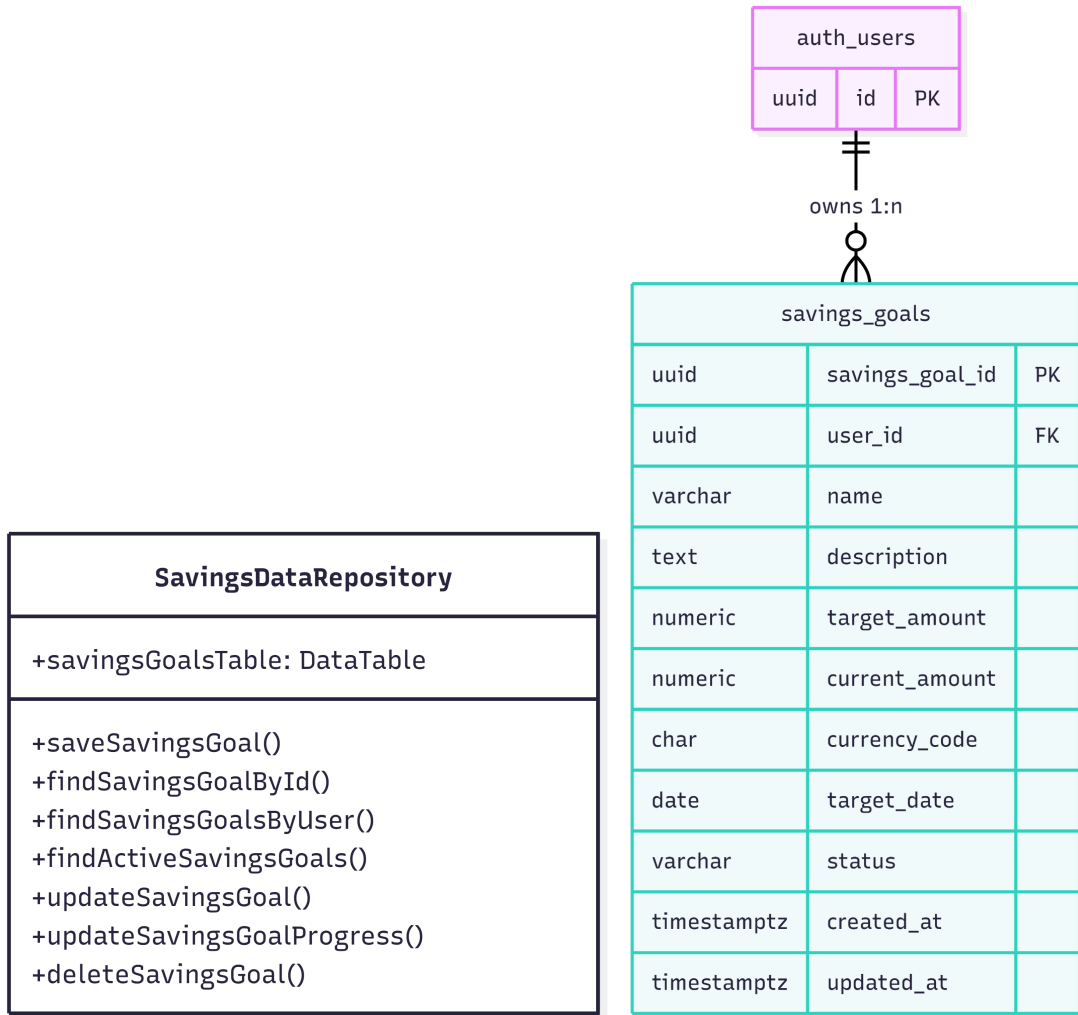
Method	Description
findRecommendationRunsByBudget	Returns all runs for a budget.
updateRecommendationRun	Updates run status and completion time.
saveRecommendedAllocation	Inserts recommended allocation.
findAllocationsByRun	Returns all allocations for a run.
deleteAllocationsByRun	Deletes all allocations for a run.
saveRecommendationHistory	Records recommendation lifecycle event.
findHistoryByRun	Returns all history for a run.
findHistoryByEventType	Returns history filtered by event type.

4.3.6 Component: Savings Data Repository

Layer: Data

Responsibility: Stores user savings goals with progress tracking.

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Method	Description
saveSavingsGoal	Inserts or updates savings goal.
findSavingsGoalById	Finds goal by savings_goal_id.
findSavingsGoalsByUser	Returns all savings goals for a user.
findActiveSavingsGoals	Returns active (non-achieved, non-cancelled) goals.

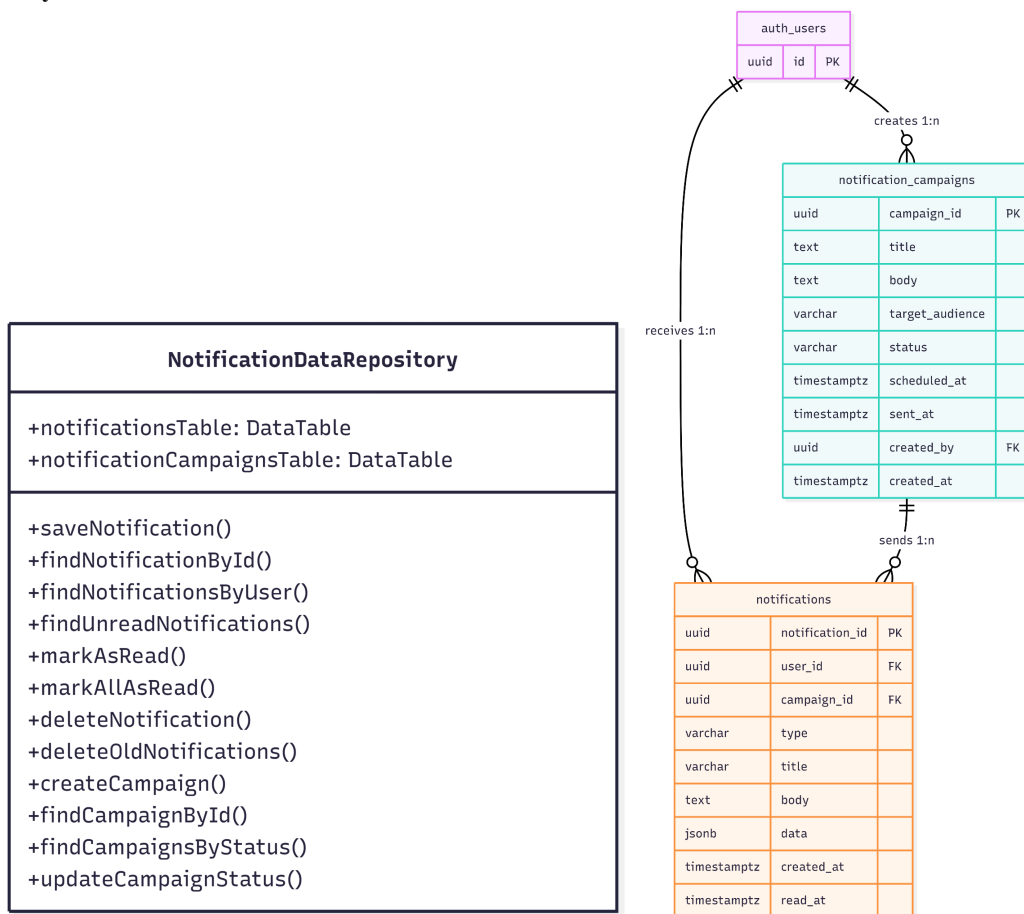
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Method	Description
updateSavingsGoal	Updates goal fields.
updateSavingsGoalProgress	Updates current_amount and status.
deleteSavingsGoal	Deletes savings goal.

4.3.7 Component: Notification Data Repository

Layer: Data

Responsibility: Stores user notifications and read status.



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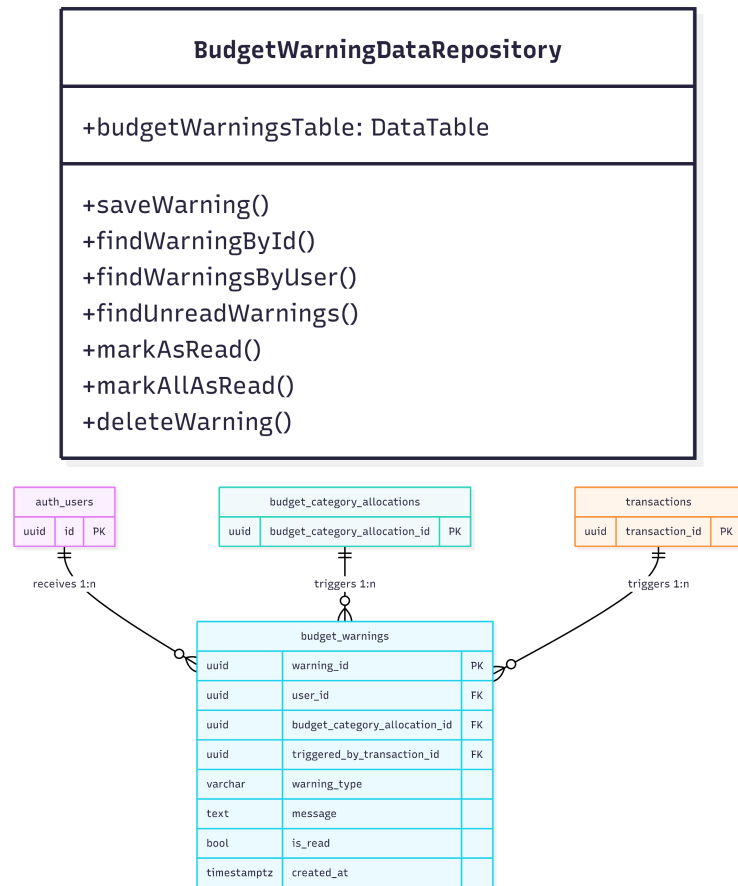
Method	Description
saveNotification	Inserts or updates notification.
findNotificationById	Finds notification by notification_id.
findNotificationsByUser	Returns all notifications for a user (paginated).
findUnreadNotifications	Returns unread notifications for a user.
markAsRead	Sets read_at timestamp.
markAllAsRead	Sets read_at for all user notifications.
deleteNotification	Deletes notification.
deleteOldNotifications	Cleans up old notifications.
createCampaign	Creates a new notification campaign.
findCampaignById	Finds campaign by campaign_id.
findCampaignsByStatus	Returns campaigns filtered by status.
updateCampaignStatus	Updates campaign status.

4.3.8 Component: Budget Warning Data Repository

Layer: Data

Responsibility: Stores budget threshold warning notifications.

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Method	Description
saveWarning	Inserts budget warning.
findWarningById	Finds warning by warning_id.
findWarningsByUser	Returns all warnings for a user.
findUnreadWarnings	Returns unread warnings for a user.
markAsRead	Sets is_read to true.

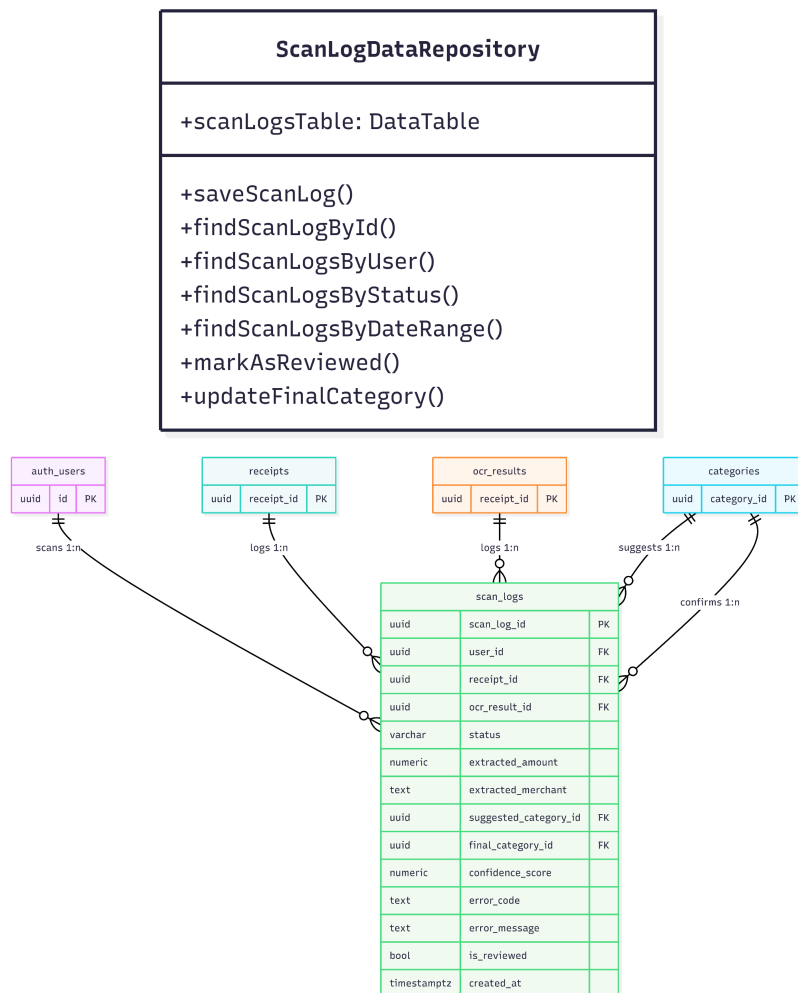
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Method	Description
markAllAsRead	Marks all user warnings as read.
deleteWarning	Deletes warning.

4.3.9 Component: Scan Log Data Repository

Layer: Data

Responsibility: Stores OCR scan audit logs for admin review and AI improvement.



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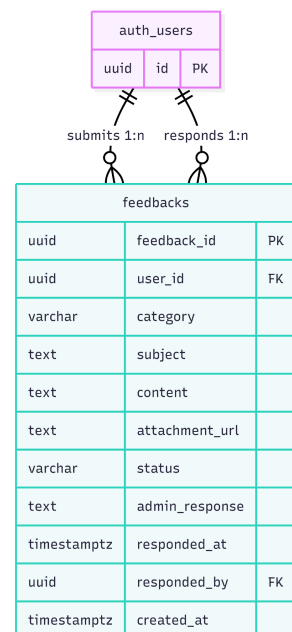
Method	Description
saveScanLog	Inserts or updates scan log.
findScanLogById	Finds scan log by scan_log_id.
findScanLogsByUser	Returns scan logs for a user.
findScanLogsByStatus	Returns scan logs filtered by status.
findScanLogsByDateRange	Returns scan logs within date range.
markAsReviewed	Marks scan log as reviewed.
updateFinalCategory	Updates final category selection.

4.3.10 Component: Feedback Data Repository

Layer: Data

Responsibility: Stores user feedback and support requests.

FeedbackDataRepository
+feedbacksTable: DataTable
+submitFeedback() +findFeedbackById() +findFeedbacksByUser() +findFeedbacksByStatus() +updateFeedbackStatus() +respondToFeedback() +getAllFeedbacks()

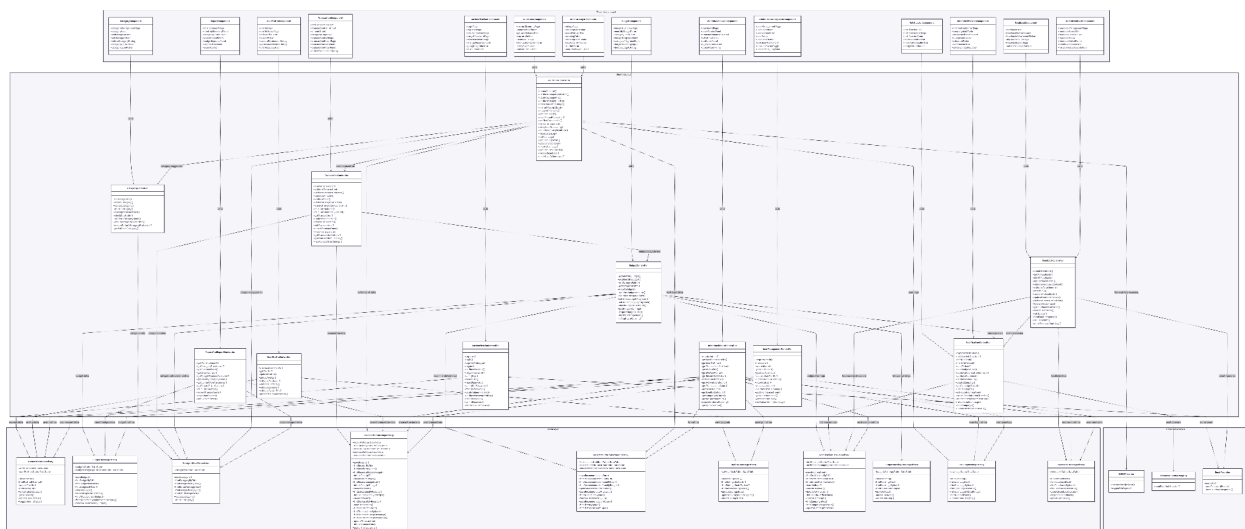


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Method	Description
submitFeedback	Submits a new feedback.
findFeedbackById	Finds feedback by feedback_id.
findFeedbacksByUser	Returns feedbacks submitted by a user.
findFeedbacksByStatus	Returns feedbacks filtered by status.
updateFeedbackStatus	Updates feedback status.
respondToFeedback	Adds admin response to feedback.
getAllFeedbacks	Returns all feedbacks (admin only).

4.4 Overall Class Diagram

The following class diagram consolidates the Presentation, Business, and Data Layer classes described in Sections 4.1, 4.2, and 4.3. Presentation classes contain UI attributes only, Business classes contain workflow methods, and Data classes contain persistence attributes and methods.

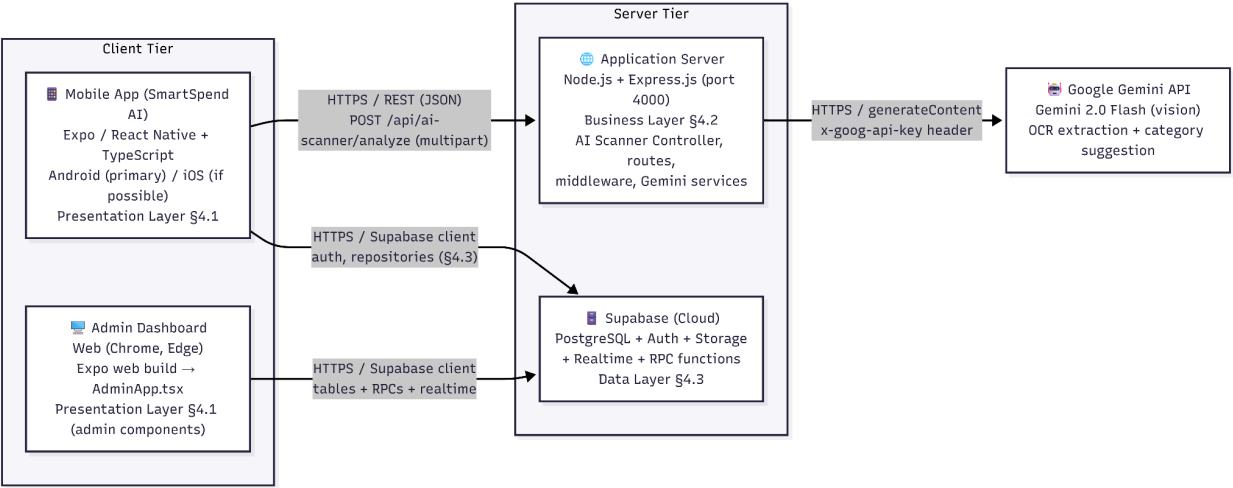


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5. Deployment

The SmartSpend AI system is deployed as a 3-tier topology that mirrors the Logical View:

- Presentation Layer runs on the client tier — as a native mobile app on user devices and as a web dashboard in the admin's browser.
- Business Layer runs on a Node.js + Express application server.
- Data Layer runs on Supabase, a cloud-hosted PostgreSQL platform that also supplies authentication, object storage, realtime, and database functions (RPCs).



5.1 Client-side

The client side consists of the SmartSpend AI mobile application and the web-based Admin Dashboard. Both are built with the same Expo / React Native (TypeScript) codebase, sharing types, state contexts, and data-layer repositories.

Mobile Application (End Users)

- **Deployment:** Built with the Expo managed workflow and deployed natively onto the user's smartphone. **Android is the primary target; iOS is deployed if possible** (per Architectural Constraints). There is no per-device installation server — the app bundle is published via Expo and installed on the device like a standard mobile application.
- **Runtime:** The mobile device runs the entire Presentation Layer — authentication screens, profile, categories, transactions, budgets, AI scanner, reports, notifications, and feedback. Business workflows not yet served by the Express backend are orchestrated client-side through React contexts (*state/*) and data repositories (*data/*).
- **Communication:**
 - **AI scanning flow:** Sends the receipt image to the Express application server via **POST /api/ai-scanner/analyze** (multipart/form-data) and receives normalized **ExtractedReceiptData**.
 - **Data flow:** Communicates directly with Supabase via the Supabase JS client (auth, transactions, categories, budgets, receipts, notifications, etc.). No server-side proxy is needed for CRUD operations.
 - **Offline resilience:** Employs local caching of manual entries with automatic synchronization to the cloud database when connectivity returns (Architectural Goal — Availability).
- **Specifications:** Optimized for Android devices of varying screen sizes and resolutions (Usability Goal, SAD §2.1). Utilizes **expo-image-picker** for camera/gallery receipt capture and **expo-notifications** for local push notifications.

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Admin Dashboard (Single Admin)

- **Deployment:** Rendered by the same Expo build as a **web application**, accessed through modern browsers (**Chrome, Edge**). The entry point `src/App.tsx` routes `/admin*` paths to `AdminApp.tsx`.
- **Communication:** Connects **directly to Supabase** to read tables (`user_profiles`, `transactions`, `scan_logs`, `ocr_results`, `feedbacks`, `audit_logs`, `notification_campaigns`, etc.), call secure **RPC functions** (`get_admin_auth_users`, `admin_update_user_profile`, `admin_update_user_account_status`, `admin_create_notification_campaign`, `admin_cancel_notification_campaign`), and subscribe to **realtime changes** for automatic metric refreshes. RBAC is enforced by the `is_admin` and `account_status` flags in `user_profiles`.
- **Specifications:** Features a responsive web layout (sidebar, metric cards, tables) and supports CSV export of user management data.

5.2 Server-side

The server side provides the Business Layer and hosts the Data Layer.

Application Server (Node.js + Express)

- **Deployment:** A Node.js process running the Express application. In development, it runs on the developer machine (`npm run dev`, port `4000`); for production, it is deployed on a Linux-based host (VPS or PaaS) behind a reverse proxy providing HTTPS, with `ALLOWED_ORIGINS` restricting CORS to known clients.
- **Responsibilities:** Handles global middleware (CORS, JSON body parsing, multer upload limits), the health endpoint (`GET /health`), and the AI Scanner routes (`/api/ai-scanner/*`) implementing the **AI Scanner Controller**. This includes `uploadReceipt`, `validateImageFile` (mime/size via multer), `processReceiptImage` (Gemini call), `extractReceiptData` / `suggestCategory` (parser service), `getScanResult`, and `listCategories`. The Gemini API key is kept **server-side only** and never exposed to the client.
- **Fault tolerance:** Utilizes a centralized error handler (`middleware/errorHandler.js`) returning structured JSON errors. On a Gemini failure, the client is redirected to the manual input form with a friendly message (Availability Goal).

Database Server (Supabase — Cloud PostgreSQL)

- **Deployment:** Managed cloud PostgreSQL hosting all tables backing the Data Layer repositories (§4.3): `categories`, `transactions`, `budgets`, `budget_category_allocations`, `receipts`, `receipt_images`, `receipt_line_items`, `user_profiles`, `ocr_results`, `recommendation_runs`, `scan_logs`, `feedbacks`, `audit_logs`, `notifications`, `notification_campaigns`, `notification_campaign_targets`, `user_notification_settings`, `budget_warnings`, and `push_tokens`.
- **Supabase Auth:** Manages email/password registration and login, OTP verification, password resets, and Google OAuth. Password hashing and session management are handled entirely by Supabase Auth.
- **Storage:** Stores receipt images and user avatars in Supabase Storage (`storage_url` columns), utilizing HTTPS in transit and at-rest encryption managed by the cloud provider.
- **Database functions (RPCs) & triggers:** Admin operations are exposed as security-definer RPCs (`get_admin_auth_users`, etc.). Scheduled helpers maintain budgets and notifications (`ensure_user_monthly_budget`, `refresh_user_budget_spending`, `evaluate_user_budget_notifications`). An `onAuthUserCreated` trigger automatically creates a profile row upon signup.
- **Realtime:** The Admin Dashboard subscribes to `postgres_changes` on dashboard tables for live metric updates.

5.3 Third-Party Services Integration — Large Language Model (Gemini)

The Large Language Model (LLM) component provides OCR and receipt auto-categorization for the AI Scanner

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Large Language Model (Gemini)

- **Deployment:** Gemini is **not hosted on SmartSpend's infrastructure**; it is accessed as an external cloud-based API over HTTPS. The AI Scanner Controller (deployed on the Express server) serves as the integration point, receiving the receipt image from the mobile client, calling Gemini, parsing the structured response, and returning normalized data.
- **Communication protocol:** HTTP-based REST API ([\[https://generativelanguage.googleapis.com/v1beta/models/gemini-2.0-flash:generateContent\]](https://generativelanguage.googleapis.com/v1beta/models/gemini-2.0-flash:generateContent)(<https://generativelanguage.googleapis.com/v1beta/models/gemini-2.0-flash:generateContent>)) using JSON payloads with the image inlined as base64, authenticated via the `x-goog-api-key` header.
- **Authentication:** API keys are stored securely in environment variables (`GEMINI_API_KEYS`) and rotated through a **key pool** (`geminiKeyPool.js`) with automatic failover when quotas are reached. Keys are never exposed to the client-side.
- **Security:** An abstraction layer (`geminiClient.js` + `receiptParser.js`) isolates all Gemini-specific logic to ensure sensitive financial data is not transmitted beyond the fields required for extraction.
- **Performance:** The AI pipeline (scan → OCR → categorization) targets a completion time of within 4.0 seconds (Efficiency Goal, SAD §2.1).
- **Fallback mechanism:** On timeout, quota exhaustion, or service unavailability, the controller returns a structured error (`GeminiApiError`). The mobile app then redirects the user to the manual input form with a meaningful message (Availability Goal, SAD §2.1), and failures are logged for the Admin AI Logs component.

Other Third-Party Services

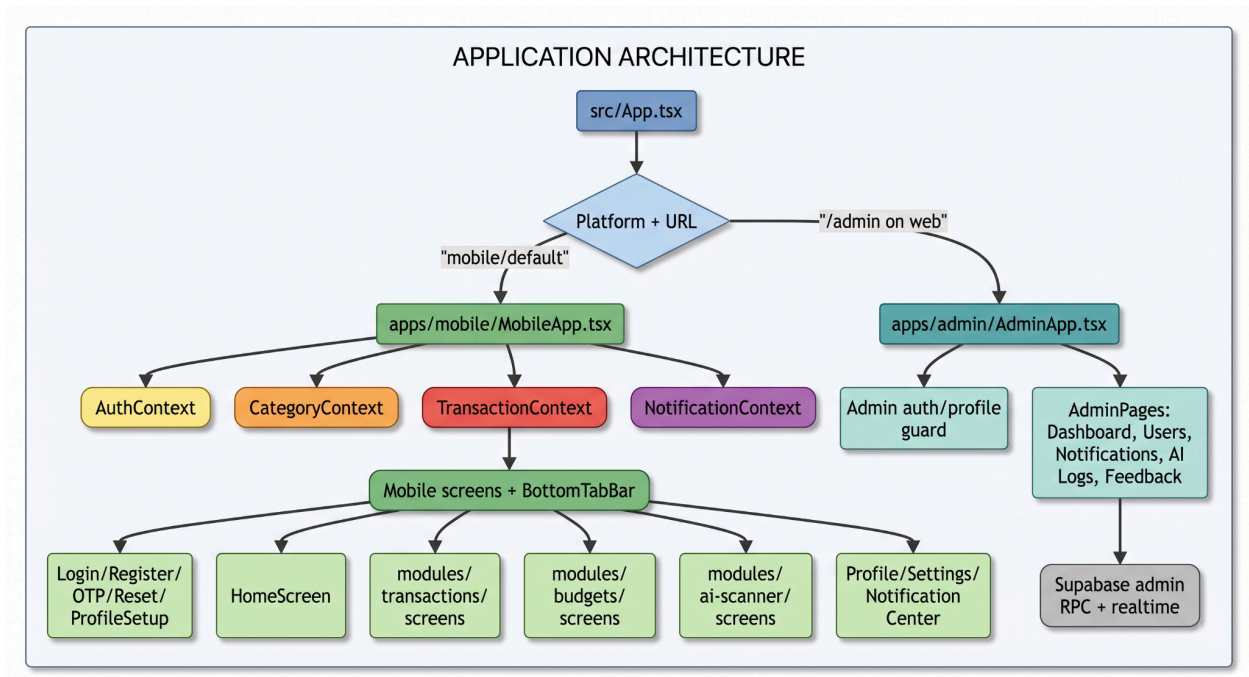
- **Supabase Auth (Google OAuth):** Google login is provided seamlessly through Supabase Auth's OAuth flow.
- **Expo Push Service (`exp.host`):** Used by the Admin Dashboard to deliver push notifications for configured campaigns (`admin_create_notification_campaign` + POST [\[https://exp.host/--api/v2/push/send\]](https://exp.host/--api/v2/push/send)(<https://exp.host/--api/v2/push/send>)).

6. Implementation View

This section describes how the Logical View components (SAD §4) are organized in the codebase. The repository is a monorepo: the client (mobile + admin web) lives under `src/`, and the Express business layer lives under `src/backend/`.

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6.1 Front-end



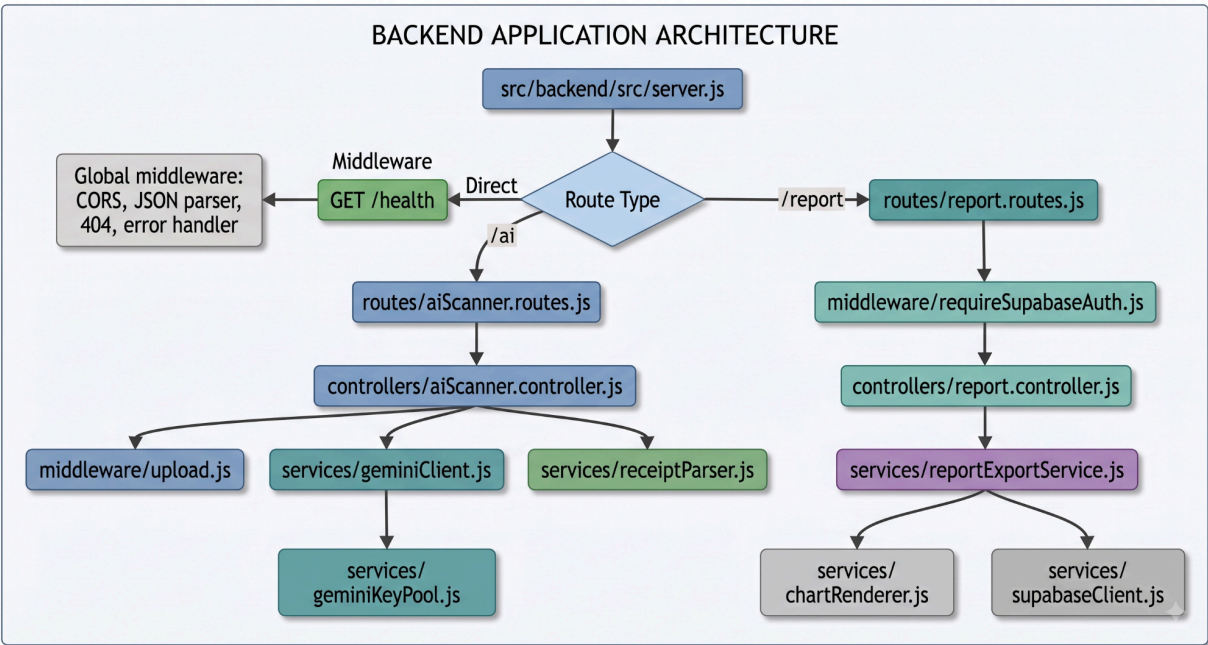
- Descriptions

Folder/file	Responsibility
App.tsx	Chooses Admin Dashboard when running on web and the URL starts with /admin; otherwise starts Mobile App.
package.json	Expo scripts: start, android, ios, web; dependencies include Expo, React Native, Supabase JS, navigation, notifications, image picker, sharing, and SVG/chart libraries.
app.json	Expo application metadata and platform configuration.
MobileApp.tsx	Mobile shell, provider composition, app-level navigation state.
screens/*	Authentication, onboarding, home, profile, settings, notification center, and placeholder screens.
components/*	Reusable mobile dashboard widgets: header, summary card, recent transactions, pie chart, category breakdown.
AdminApp.tsx	Admin shell, role/account validation, data loading, realtime subscriptions, dashboard actions, modal orchestration.

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pages/DashboardPage.tsx	System/user/AI/transaction metrics and warning panels.
pages/UsersPage.tsx	User search, filter, detail, edit, status lock/unlock, CSV export.
pages/NotificationsPage.tsx	Notification campaign list, create/open/cancel/delete campaign, emitted user notifications.
pages/AILogsPage.tsx	AI scan log list, confidence/latency metrics, review filters.
pages/FeedbackPage.tsx	Feedback queue, status/type filtering, admin processing entry.

6.2 Back-end



- Descriptions:

Folder/file	Responsibility
server.js	Express app entry, CORS, JSON body parser, health endpoint, route mounting, centralized 404/error handling.
routes/aiScanner.routes.js	AI scanner HTTP routes.
controllers/aiScanner.controller.js	Upload validation, Gemini invocation, receipt parsing, category suggestion, scanner responses.
routes/report.routes.js	Report/export HTTP routes.

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controllers/report.controller.js	Report request validation and response handling.
middleware/upload.js	Multipart receipt image upload constraints.
middleware/requireSupabaseAuth.js	Validates Bearer Supabase token for protected backend routes.
middleware/errorHandler.js	Central JSON error response.
services/geminiClient.js	Calls Gemini OCR/LLM endpoint.
services/geminiKeyPool.js	Rotates Gemini API keys and handles quota/failure selection.
services/receiptParser.js	Converts raw LLM output into normalized receipt data and category suggestions.
services/reportExportService.js	Builds export data/files from Supabase data.
services/chartRenderer.js	Renders simple chart assets for report exports.
services/supabaseClient.js	Backend-side Supabase client.